Unit One

Dynamic Electricity and Electromagnetism

Chapter One	Electrical Current and Ohm's Law.
Chapter Two	Magnetic Effects of the Electric Current and Measuring Instruments.
Chapter Three	Electromagnetic Induction.
Chapter Four	Alternating Current Circuits.



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CHAPTER

ONE

Electrical Current and Ohm's Law

Lesson 1 Electrical Current and Ohm's Law.

Lesson 2 Resistors Connection.

Lesson 3 Ohm's Law for the Closed Circuit.

Lesson 4 Kirchhoff's Laws.

- Questions on each lesson from the Previous Years Exams.
- Test on Chapter 1





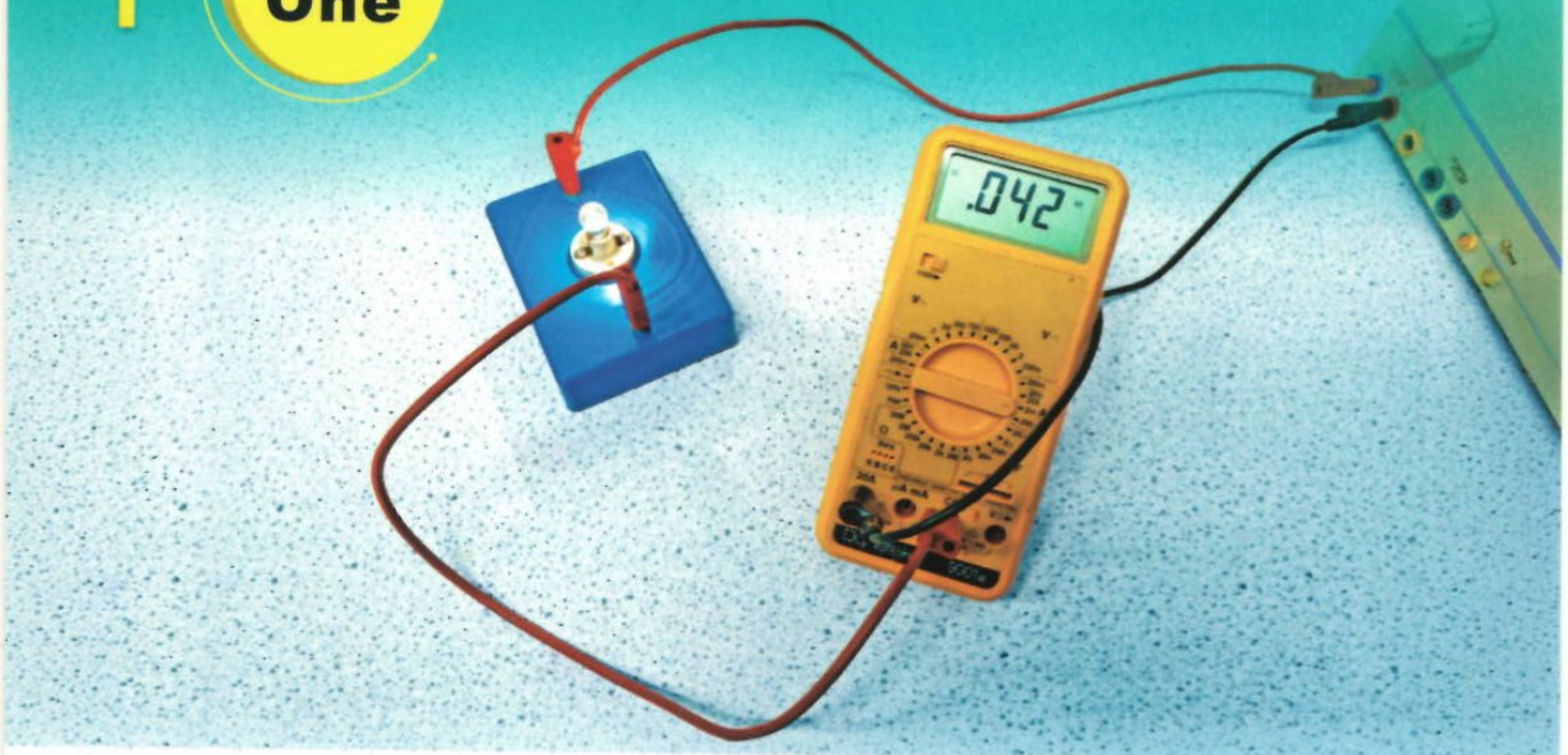
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Chapter 1

Questions on Lesson One

Electrical Current and Ohm's Law

Interactive test



The questions signed by ✱ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

Electric current intensity (I)

=

$\frac{Q}{t}$ $\frac{V}{R}$ $\frac{Ne}{t}$

- To determine the number of the passing electrons through a cross-section of a conductor (N): $N = \frac{Q}{e}$ (electron)

Electric potential difference (V)

=

$\frac{W}{Q}$ IR $\frac{W}{Ne}$

- To determine the resistivity (ρ_e) and the conductivity (σ):

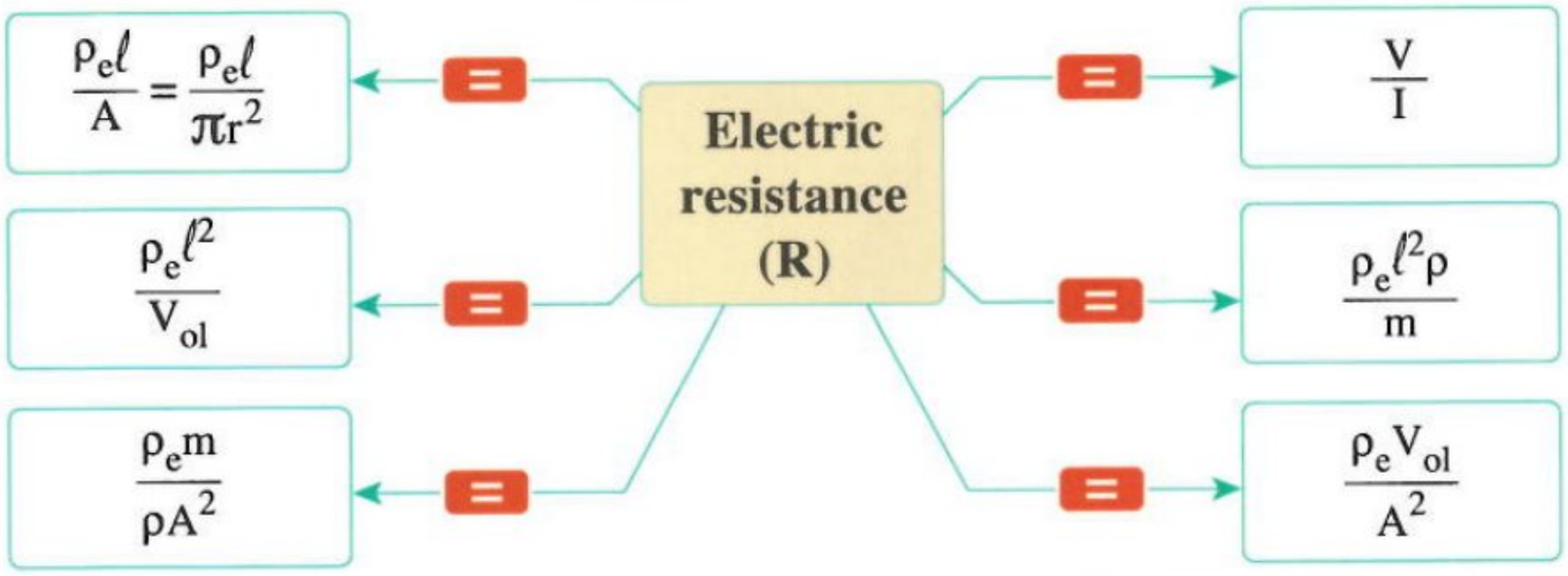
$\rho_e = \frac{RA}{\ell}$ (Ω.m) $\quad \therefore \sigma = \frac{1}{\rho_e} = \frac{\ell}{RA}$ ($\Omega^{-1}.m^{-1}$)

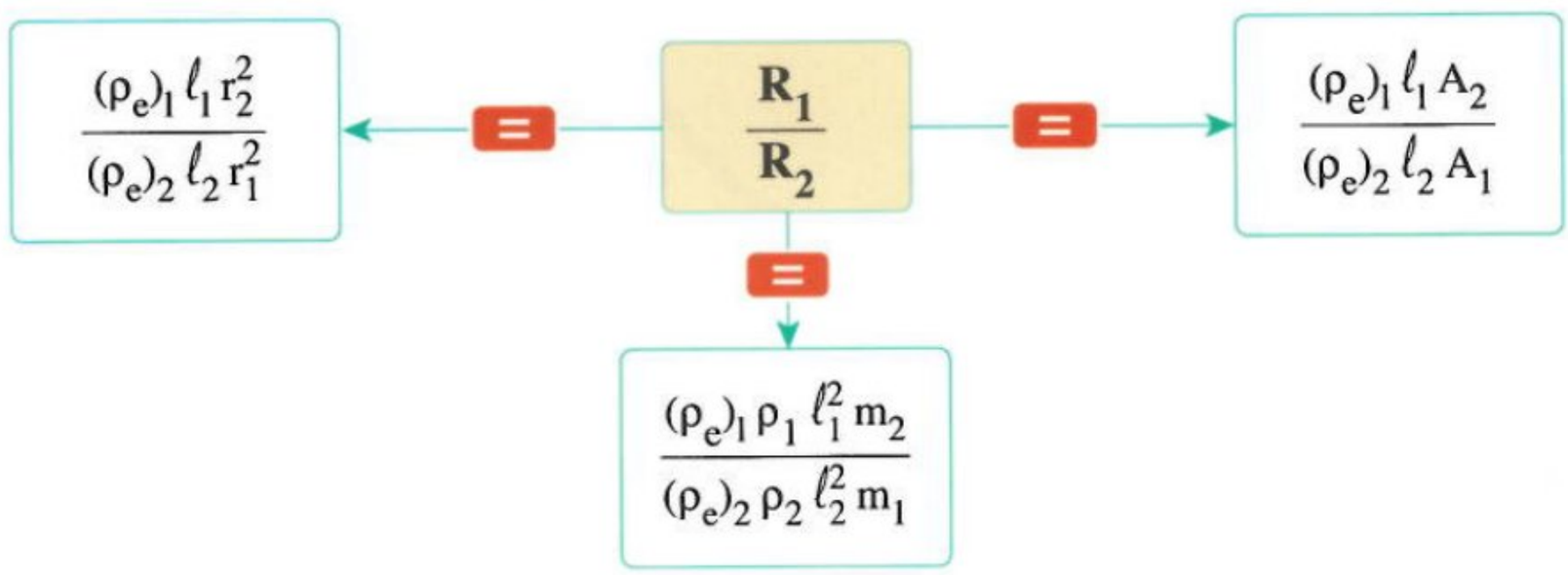
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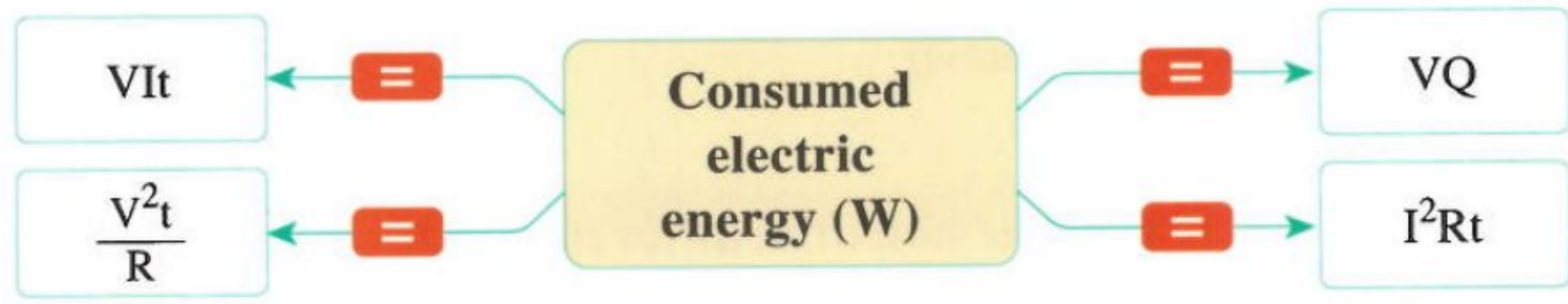
- To determine the length of a wire that is coiled to form a circular coil of number of turns N and radius r_{coil}: $\ell_{wire} = 2\pi r_{coil} N$

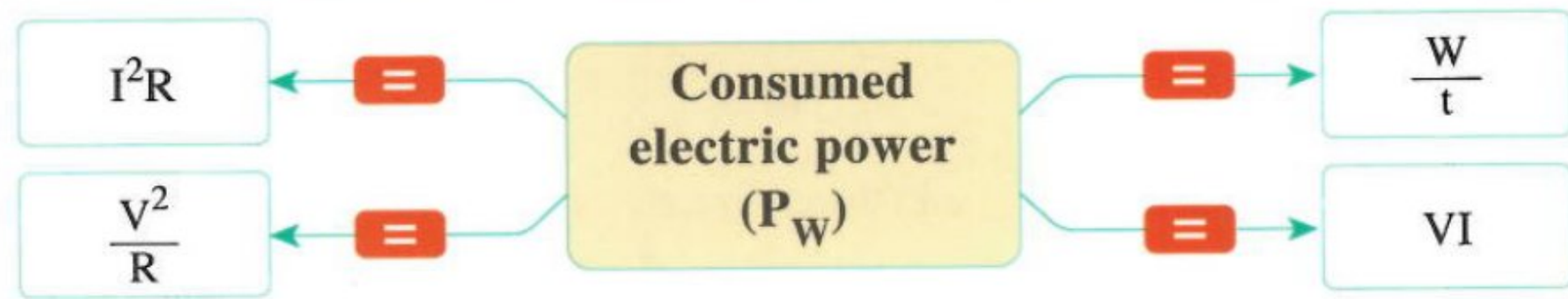




- If the wire is reshaped so that its length increases and its cross-sectional area decreases, so:

$$\frac{R_1}{R_2} = \frac{\ell_1 A_2}{\ell_2 A_1} = \frac{\ell_1^2}{\ell_2^2} = \frac{A_2^2}{A_1^2} = \frac{r_2^4}{r_1^4}$$





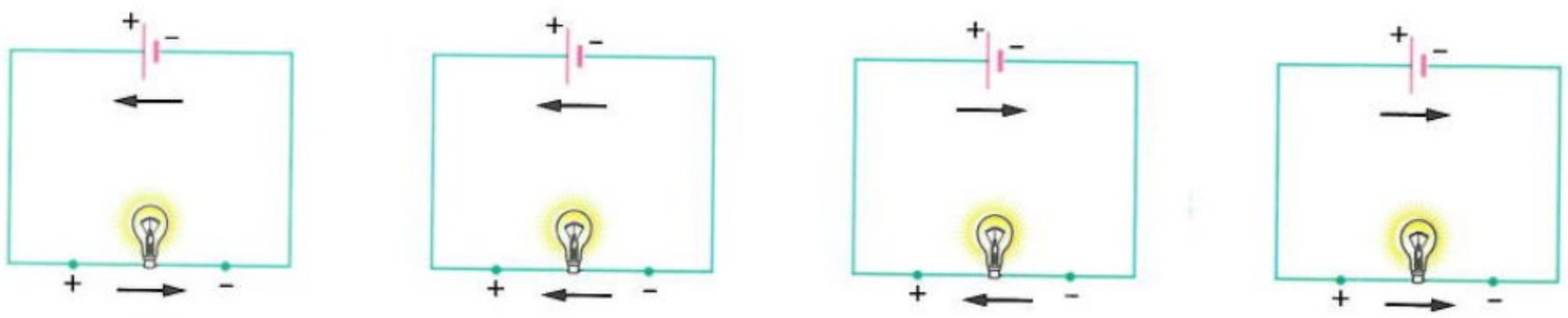
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First Multiple choice questions

Use the following constant when needed: $e = 1.6 \times 10^{-19}$ C

Electric current intensity

1 In which of the following electric circuits do the directions of arrows correctly represent the conventional direction of electric current?



(a) (b) (c) (d)

2 The opposite figure represents a section of an electric circuit through which a direct electric current (I) is passing, so which of the two ammeters shows a higher reading?



(a) Ammeter A_1 (b) Ammeter A_2

(c) Both ammeters show the same reading. (d) The answer is indeterminable.

3 If the electric current through a conductor equals 2 A, it means that:

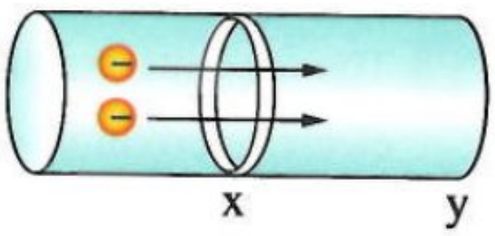
(i) the quantity of electric charge that passes through a cross-section of the conductor in 10 s is ..... .

(a) 0.2 C (b) 8 C (c) 12 C (d) 20 C

(ii) the time taken by an electric charge of 4 mC to pass through a cross-section of the conductor is ..... .

(a) 2 ms (b) 4 ms (c) 6 ms (d) 8 ms

4 In the opposite figure, a negative electric charge of 2 μC passes through cross-section x of a conductor in 4 ms, then the intensity and the conventional direction of the current are ..... , ...



(a) 8 mA, from x to y (b) 8 mA, from y to x

(c) 0.5 mA, from x to y (d) 0.5 mA, from y to x

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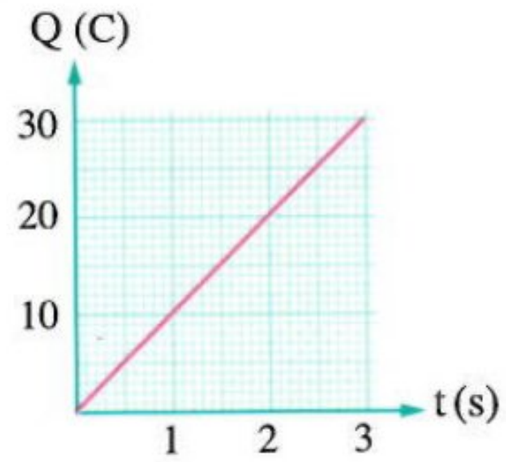
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5 If the flow rate of free electrons in wire x is 10^{20} electrons/s and its flow rate in wire y is 2×10^{20} electrons/s, then the ratio between the current intensities in the two wires $(\frac{I_x}{I_y})$ equals ..... .

(a) $\frac{1}{1}$ (b) $\frac{2}{1}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

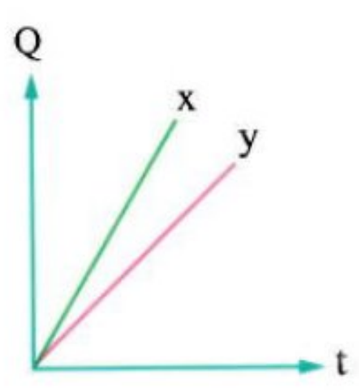
6 The opposite graph represents the relation between the quantity of electric charge (Q) that flows through a cross-section of a conductor in a DC (direct current) circuit and the time (t), so the intensity of the direct current is ..... .



(a) 2 A (b) 10 A

(c) 50 A (d) 250 A

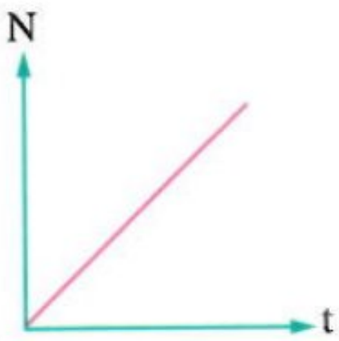
7 The opposite graph represents the relation between the electric charge (Q) passing through cross-sections of two conductors x and y and time (t), so the ratio between the electric current intensities in the two conductors $(\frac{I_x}{I_y})$ is ..... .

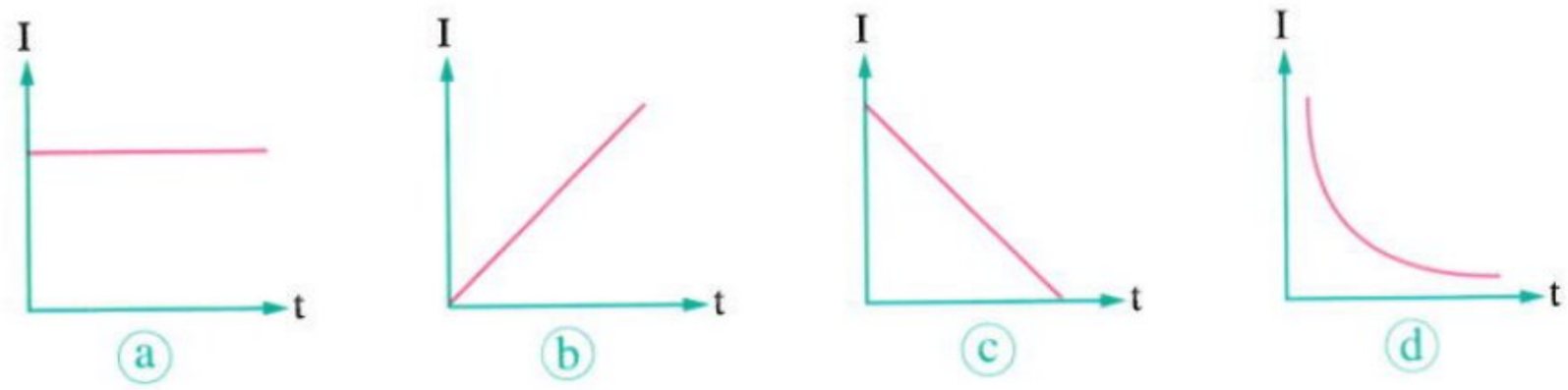


(a) greater than one (b) less than one

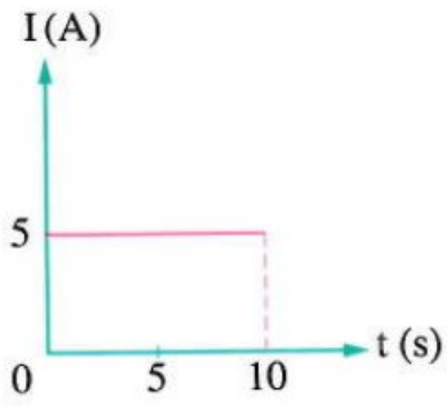
(c) equal to one (d) indeterminable

8 The opposite graph represents the relation between the number of electrons (N) that pass through a certain cross-section of a conductor in an electric circuit and the time (t), so the graph that represents the relation between the current intensity (I) through this conductor and the time (t) is ..... .





9 The opposite graph represents the relation between the electric current intensity (I) passing in a conductor and the elapsed time (t), then the electric charge that passes through a cross-section of that conductor in 10 s equals ..... .

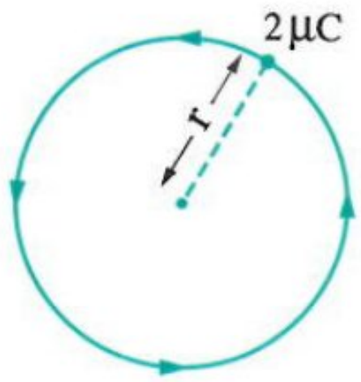


(a) 5 C (b) 10 C

(c) 15 C (d) 50 C

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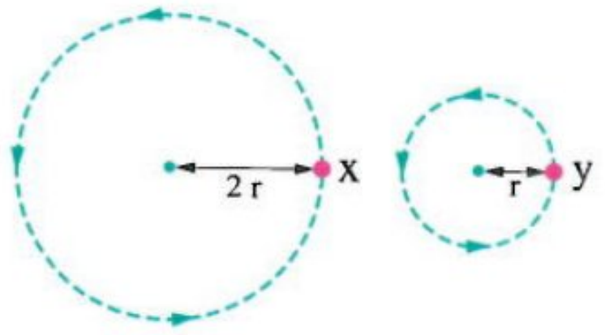
10 The opposite figure represents a charge of 2 μC rotating with a frequency of 1000 Hz in a uniform circular path, then the current that arises due to the rotation of the charge equals ..... .



(a) 5 mA (b) 2 mA

(c) 500 A (d) 2000 A

11 In the opposite figures, two bodies x and y are carrying electric charges of quantities Q and 2Q respectively, while rotating in two different circular paths, so the ratio between the current intensities due to the rotations of these charges $\left(\frac{I_x}{I_y}\right)$ if both charges rotate with:



(i) the same frequency equals ..... .

(a) $\frac{1}{1}$ (b) $\frac{1}{2}$ (c) $\frac{2}{1}$ (d) $\frac{1}{4}$

(ii) the same linear speed equals ..... .

(a) $\frac{1}{1}$ (b) $\frac{1}{2}$ (c) $\frac{2}{1}$ (d) $\frac{1}{4}$

12 Based on Bohr's model for the hydrogen atom, the electron moves in a circular path of radius 0.53 Å with a speed of 2.2×10^6 m/s, then the current intensity due to the motion of the electron equals ..... .

(a) 3.141 mA (b) 6.282 mA (c) 1.166 mA (d) 1.057 mA

Electric potential difference

13 The battery in an electric circuit is the source of ..... .

(a) positive charges (b) negative charges

(c) electric conductivity (d) electric energy

14 A metal conductor has its terminals connected to a DC source forming a closed electric circuit, so which of the following statements correctly describes the motion of free electrons inside and outside the source?

(a) They vibrate in their positions without moving to new positions.

(b) They move inside the source towards the terminal of higher potential.

(c) They move outside the source towards the terminal of lower potential.

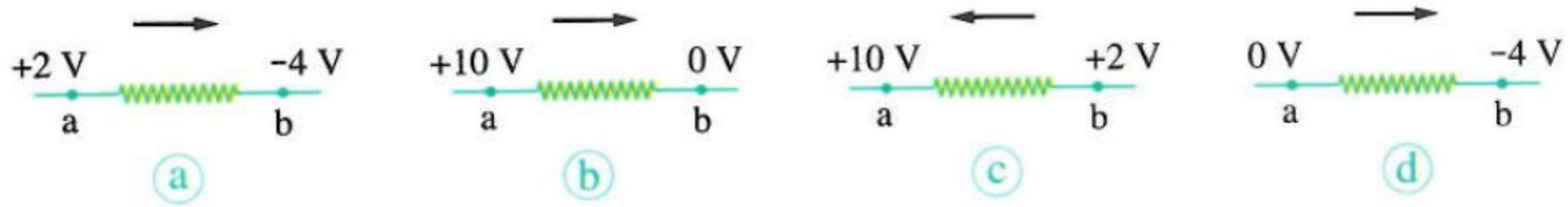
(d) They move outside the source towards the terminal of higher potential.

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15 In which of the following cases does the arrow represent the direction of motion of free electrons through the resistor between the two points a and b?



16 In the opposite figure, the work done to move a charge of 4 C between the two points x and y is ..... .



(a) 20 J (b) 40 J (c) 60 J (d) 80 J

17 The potential difference between two points when work of 30 J is required to transfer an electric charge of 10 C between them equals ..... .

(a) 0.3 V (b) 3 V (c) 30 V (d) 300 V

18 * If the work done to transfer a quantity of electric charge of 5 C between two points in a conductor is 100 J and this quantity passes through a cross-section of the conductor during 1 s, then:

(i) The potential difference between the two points equals ..... .

(a) 0.05 V (b) 5 V (c) 10 V (d) 20 V

(ii) The electric current intensity through the conductor equals ..... .

(a) 2.5 A (b) 5 A (c) 7 A (d) 12 A

(iii) The number of free electrons that pass through a cross-section of the conductor during 2 s equals ..... .

(a) 4.22×10^{18} electrons (b) 1.56×10^{19} electrons

(c) 6.25×10^{19} electrons (d) 1.25×10^{19} electrons

19 The coulomb equals the quantity of electric charge that ..... .

(a) when it passes through a cross-section of a conductor in 5 s, it means that the intensity of the current passing in the conductor is 50 A

(b) when it passes through a cross-section of a conductor in 50 s, it means that the intensity of the current passing in the conductor is 0.5 A

(c) requires a work done of 5 J to be transferred between two points of potential difference 0.5 V

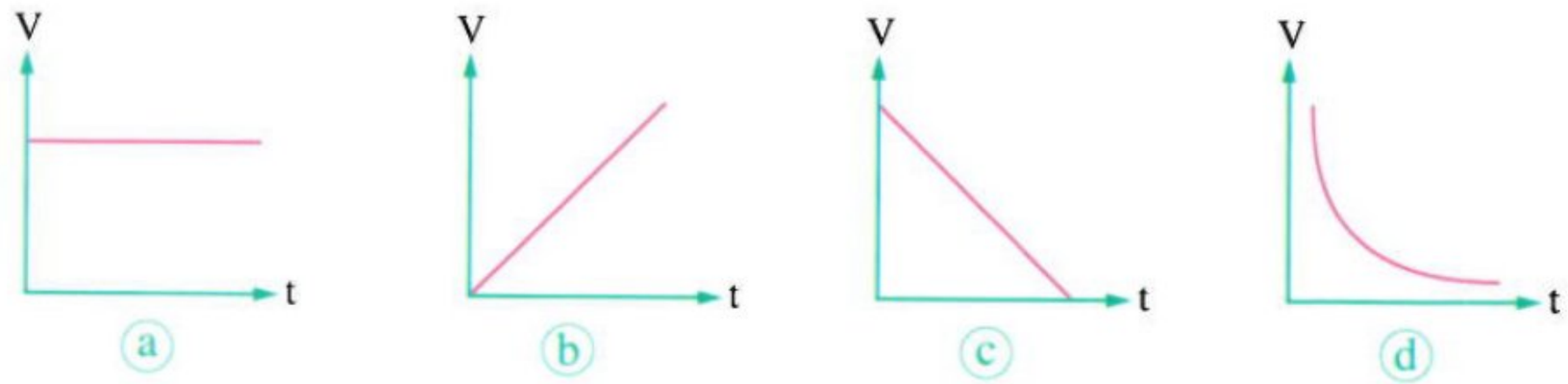
(d) requires a work done of 0.05 J to be transferred between two points of potential difference 0.05 V

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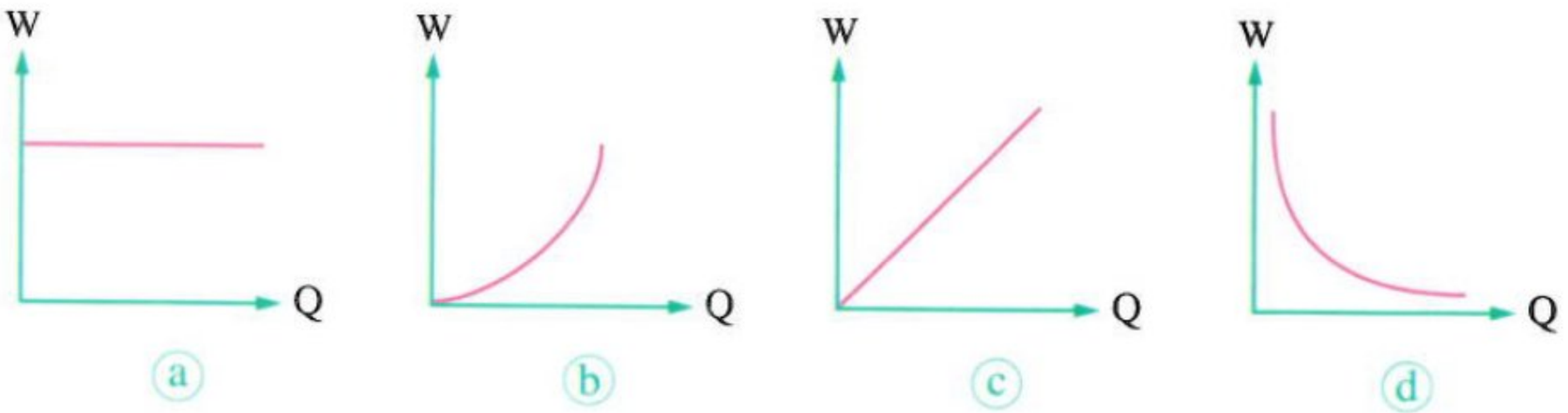
20 If the work done to transfer 10^{20} electrons between the terminals of conductor A is 10 J and the work done to transfer 4×10^{20} electrons between the terminals of conductor B is 20 J, then the ratio between the potential differences across the two conductors $(\frac{V_A}{V_B})$ equals ..... .

(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{1}{4}$ (d) $\frac{4}{1}$

21 Which of the following graphs represents the relation between the potential difference (V) across the terminals of a conductor that carries a direct current (DC) and the time (t)?

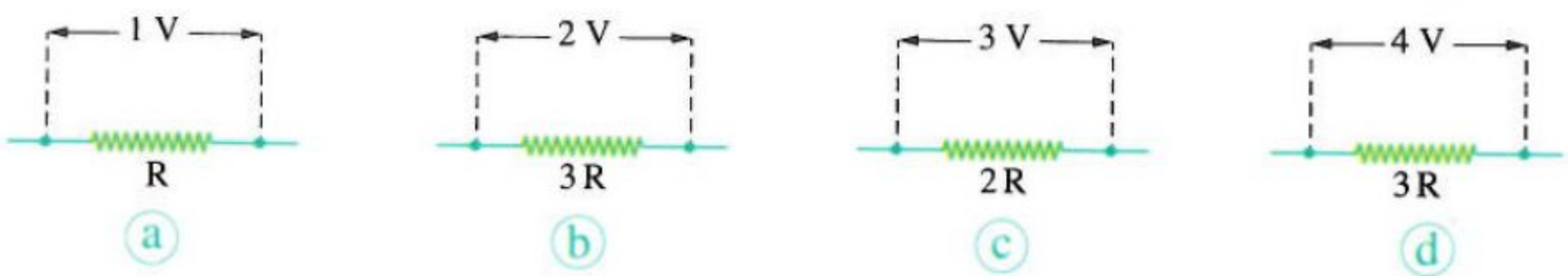


22 Which of the following graphs represents the relation between the work done (W) and the transferred electric charge (Q) between two points across which the potential difference is constant?

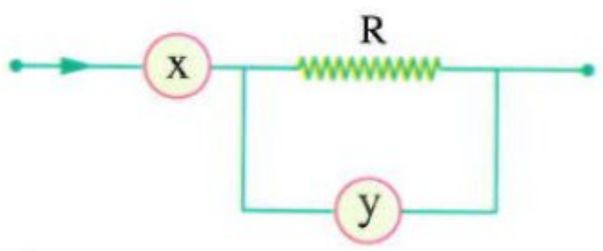


Ohm's law

23 In which of the following cases is the intensity of the current in the resistor higher?



24 The opposite diagram represents a part of an electric circuit, so if devices x and y are connected correctly, then:



(i) The two devices are ....., ... respectively.

(a) ammeter, ammeter (b) ammeter, voltmeter

(c) voltmeter, ammeter (d) voltmeter, voltmeter

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(ii) The measuring units for the measured quantities by device x and device y are ..... , ... respectively.

(a) coulomb/second, volt (b) coulomb/second, ampere

(c) joule/coulomb, volt (d) joule/coulomb, ampere

25 A potential difference of 12 V is applied between the terminals of a conductor to move 6.25×10^{18} electrons through a cross-section of this conductor during two seconds, then the resistance of the conductor is ..... .

(a) 24 Ω (b) 12 Ω (c) 6 Ω (d) 3.84 Ω

26 The opposite figures (1) and (2) show two parts of two electrical circuits carrying currents I_1 and I_2 respectively, so the ratio between current intensities $\left(\frac{I_1}{I_2}\right)$ is equal to ..... .



Figure (1)

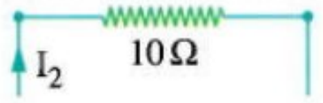


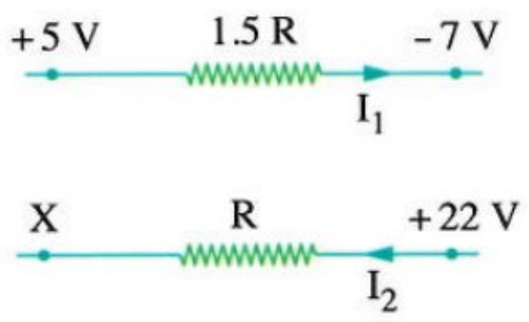
Figure (2)

(a) $\frac{4}{1}$ (b) $\frac{1}{4}$ (c) $\frac{8}{5}$ (d) $\frac{5}{8}$

27 A conductor of resistance 10 Ω carries an electric current of intensity 0.5 A. If an electric current of intensity 1 A passes through the same conductor at constant temperature, then its resistance equals ..... .

(a) 2.5 Ω (b) 5 Ω (c) 10 Ω (d) 20 Ω

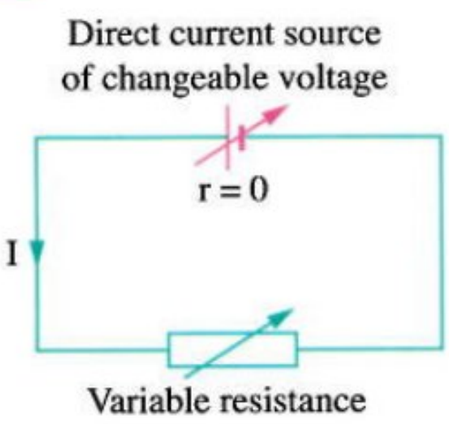
28 In the opposite figure two conductors, each carries an electric current, so if the ratio between the electric current intensities $\left(\frac{I_1}{I_2}\right)$ equals $\frac{1}{3}$, the potential of point X is ..... .



(a) 2 V (b) 24 V

(c) – 2 V (d) – 24 V

29 The opposite figure illustrates an electric circuit carrying an electric current of intensity I. One change can be made to the circuit as follows:



(I) Increasing the electromotive force of the battery to the double.

(II) Decreasing the electromotive force of the battery to its half.

(III) Increasing the variable resistance to the double.

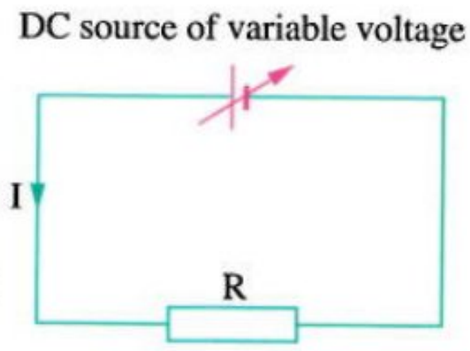
(IV) Decreasing the variable resistance to its half.

So, which of these changes will lead to increasing the electric current intensity to 2I?

(a) (I) or (II). (b) (II) or (III). (c) (III) or (IV). (d) (I) or (IV).

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30 In the opposite electric circuit, when the voltage of the source increases, the physical quantity whose value **doesn't** change is ..... .



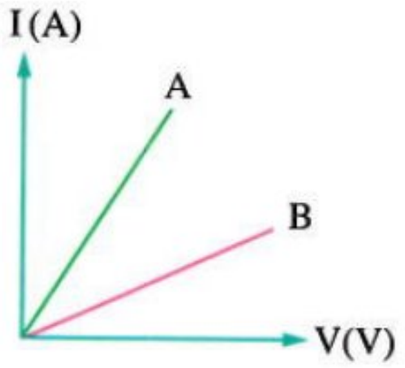
(a) the rate of passing electrons through a cross-section of the circuit

(b) the work done by the source per second

(c) the current intensity (I) passing in the circuit

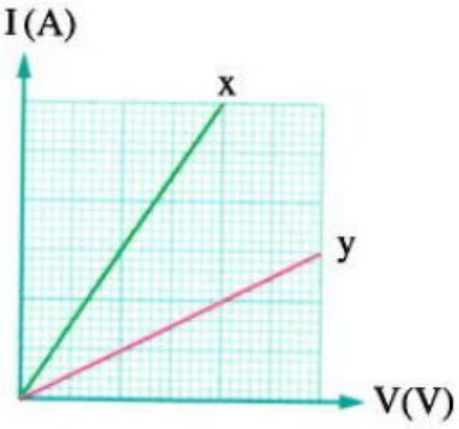
(d) the value of resistance (R)

31 The opposite graph represents the relation between the current intensity through each of two wires A and B and the potential difference across each of them, then which of the two wires has a higher resistance? And why?



	The wire of the higher resistance	The reason is that the slope of the line represents
(a)	A	the resistance of the wire
(b)	A	the reciprocal of the resistance of the wire
(c)	B	the resistance of the wire
(d)	B	the reciprocal of the resistance of the wire

32 The opposite graph represents the relation between the current intensity (I) passing through two conductors x, y and the potential difference (V) across each of them, then the ratio between the resistances of the two conductors $\left(\frac{R_x}{R_y}\right)$ equals ..... .

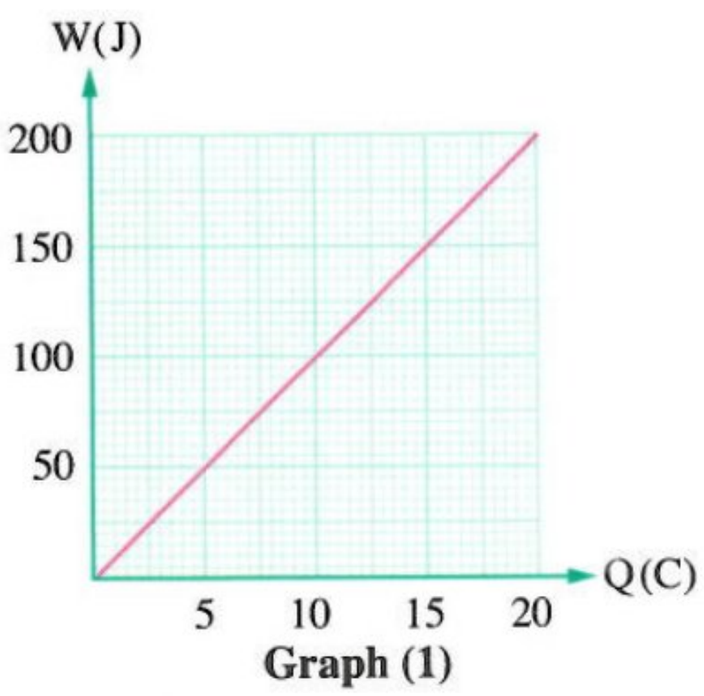


(a) $\frac{1}{2}$ (b) $\frac{3}{4}$

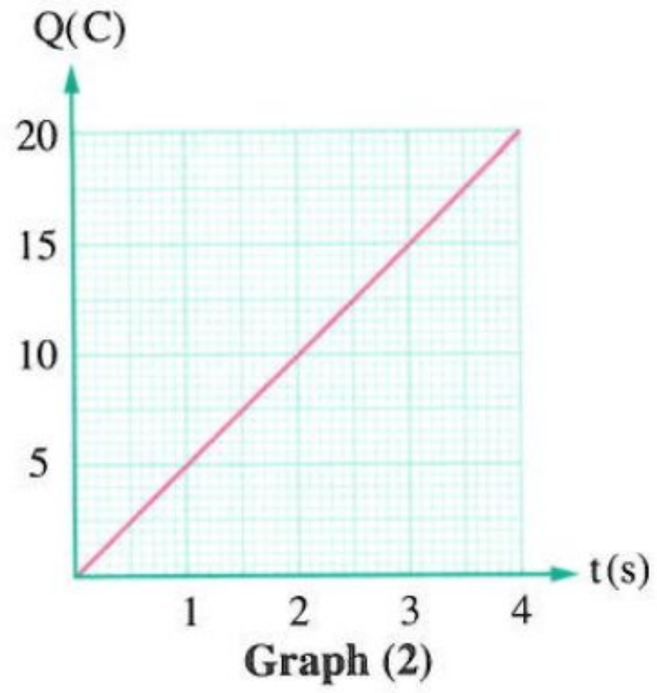
(c) $\frac{7}{4}$ (d) $\frac{1}{3}$

33 Graph (1) represents the relation between the work done (W) to move an electric charge through a conductor that carries an electric current and the amount of that charge (Q) and graph (2) represents the relation between the amount of electric charge (Q) passing through a specific cross-section of the same conductor and time (t),

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Graph (1)

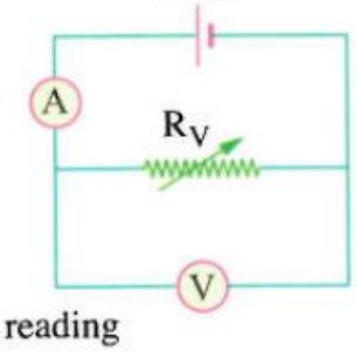


Graph (2)

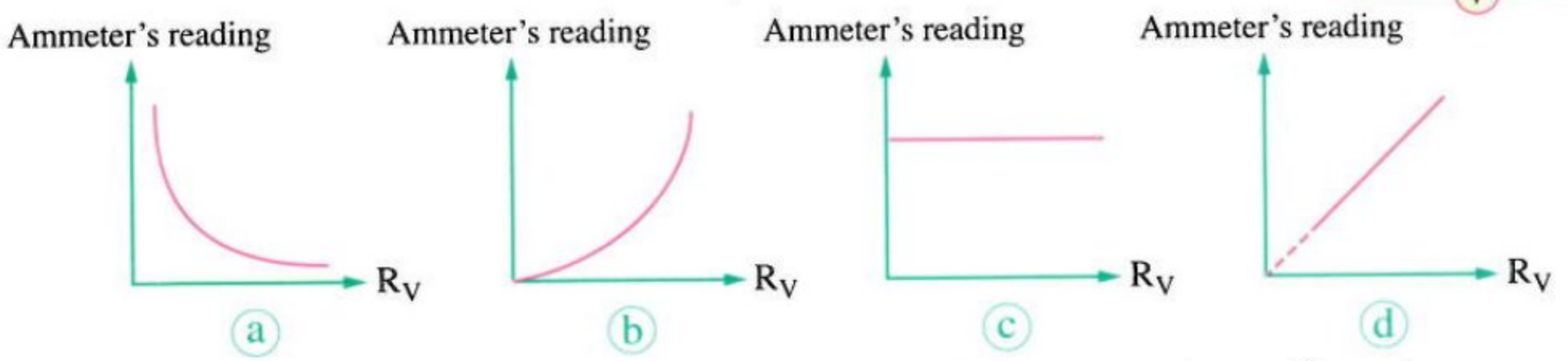
So, the electrical resistance of the conductor equals ..... .

(a) 0.5 Ω (b) 2 Ω (c) 5 Ω (d) 15 Ω

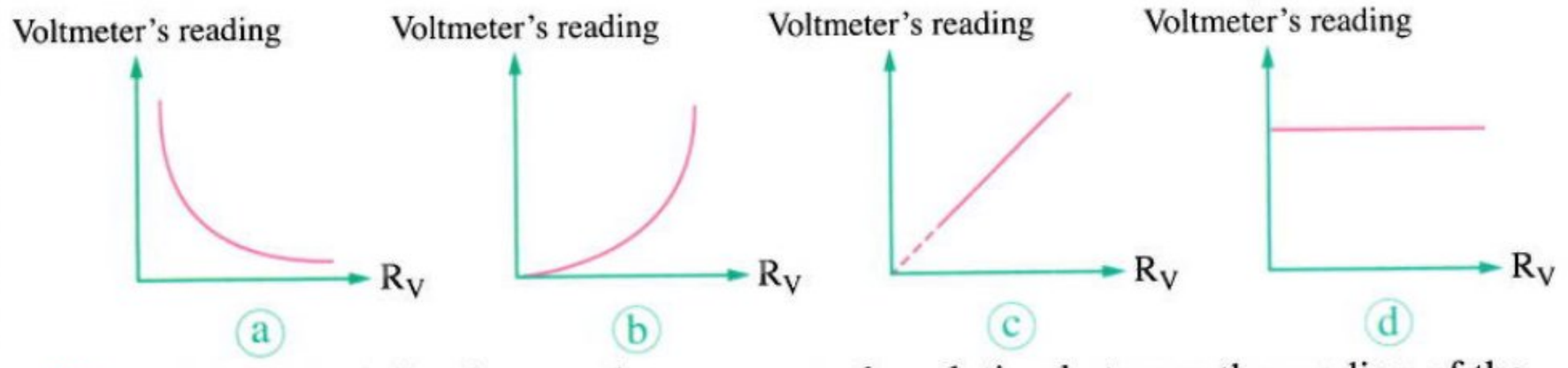
34 In the opposite circuit:



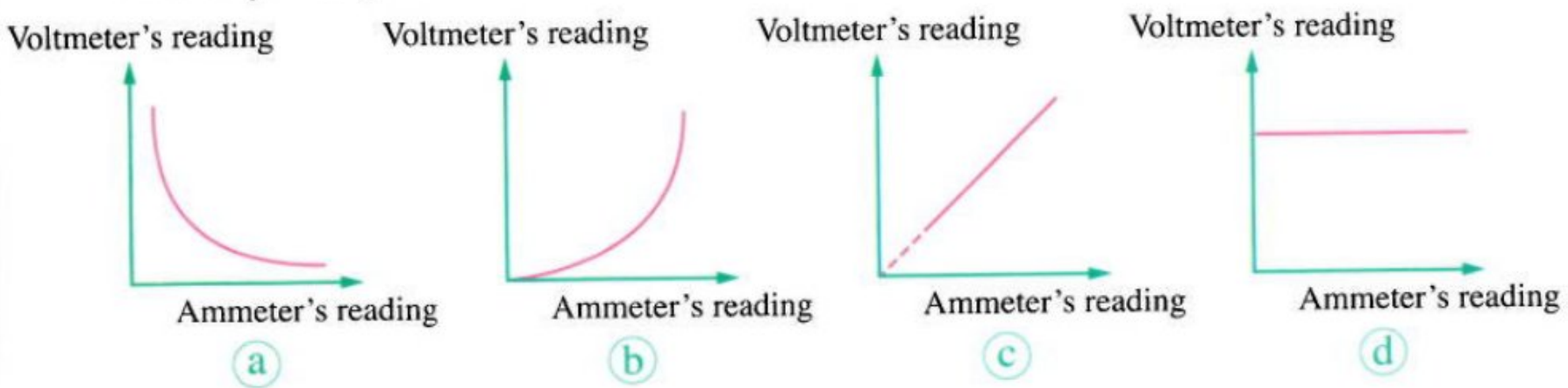
(i) Which of the following graphs represents the relation between the reading of the ammeter and the value of the resistance which is taken from R_V?



(ii) Which of the following graphs represents the relation between the reading of the voltmeter and the value of the resistance which is taken from R_V?



(iii) Which of the following graphs represents the relation between the reading of the voltmeter and that of the ammeter when the value of the resistance which is taken from R_V changes?



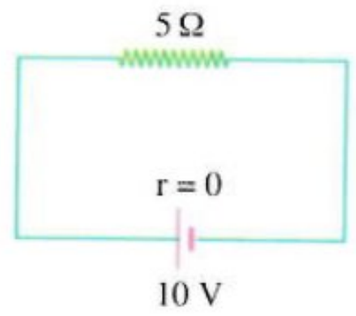
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Electric power and electric energy

35 Which of the following units **isn't** equivalent to watt?

(a) J/s (b) A.V (c) $A^2.\Omega$ (d) $\Omega^2.V$

36 In the opposite circuit:



(i) The consumed power in the circuit is ..... .

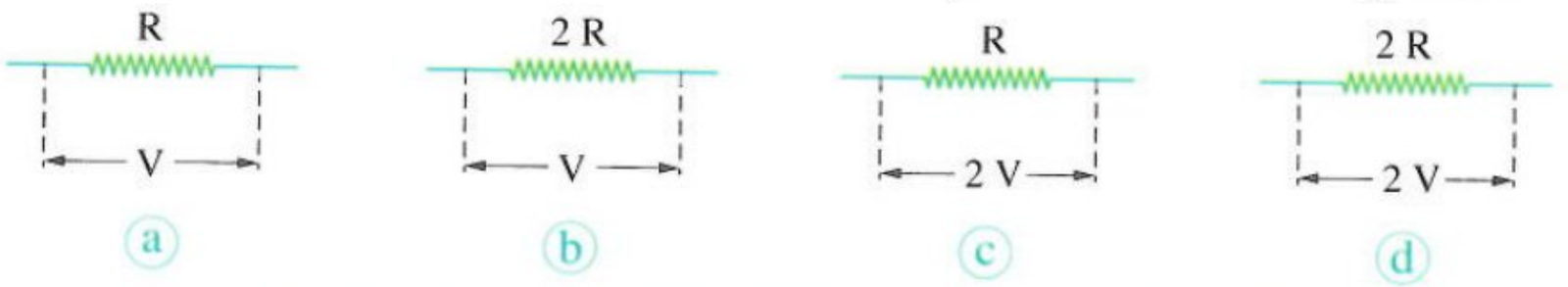
(a) 10 W (b) 20 W

(c) 50 W (d) 100 W

(ii) The consumed energy in the circuit within 20 s is ..... .

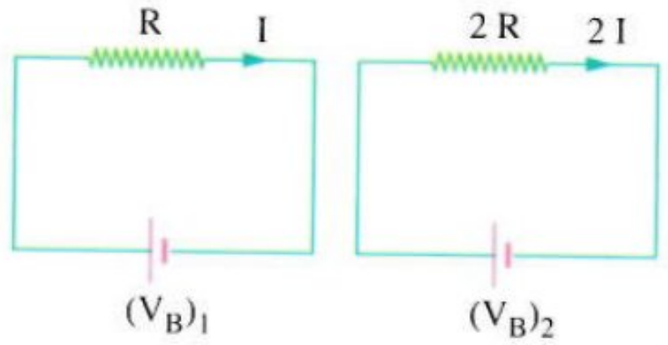
(a) 100 J (b) 200 J (c) 400 J (d) 1000 J

37 In which of the following figures is the consumed power in the resistor greater?



(a) (b) (c) (d)

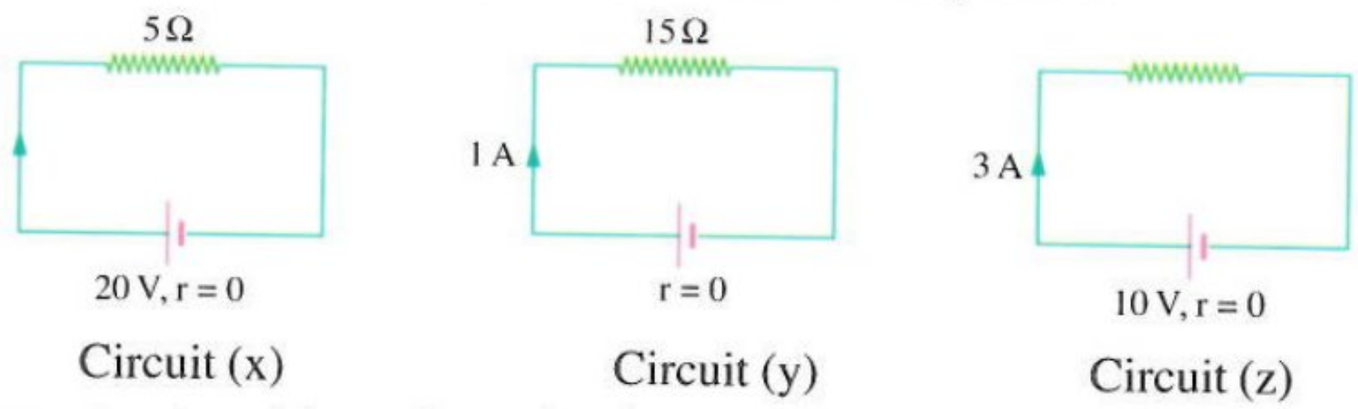
38 In the opposite two circuits, the ratio between the consumed power in the two circuits $\left(\frac{(P_w)_2}{(P_w)_1}\right)$ is ..... .



(a) $\frac{2}{1}$ (b) $\frac{4}{1}$

(c) $\frac{8}{1}$ (d) $\frac{16}{1}$

39 The following figures represent three electric circuits x, y and z.



Circuit (x) Circuit (y) Circuit (z)

So, the correct order of these three circuits according to the consumed power from the source is .....

(a) x < y < z (b) z < y < x (c) x < z < y (d) y < z < x

40 An electric fan is labelled (220 V – 100 W) and an electric heater is labelled (220 V – 1000 W), then the resistance of the heater is ..... that of the fan.

(a) equal to (b) less than

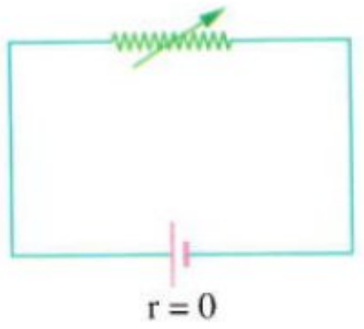
(c) greater than (d) indeterminable with respect to

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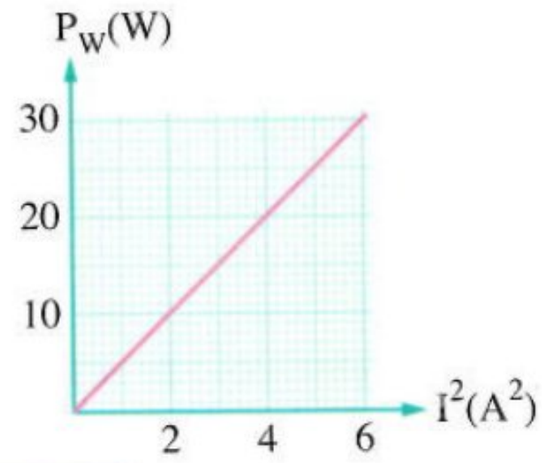
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41 In the opposite electrical circuit, when the value of the variable resistance is doubled, the rate of electrical energy consumption in the circuit ..... .



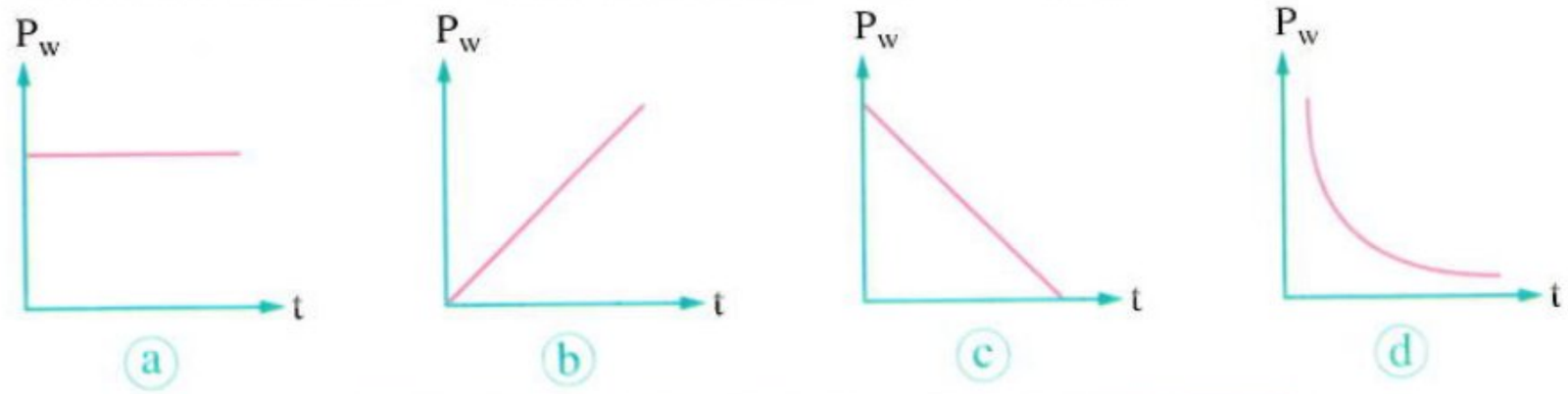
(a) increases to double
(b) decreases to its half
(c) decreases to its quarter
(d) remains constant

42 The opposite graph represents the relation between the consumed power in a conductor and the square of the intensity of the electric current that passes through it, so the resistance of the conductor equals ..... .

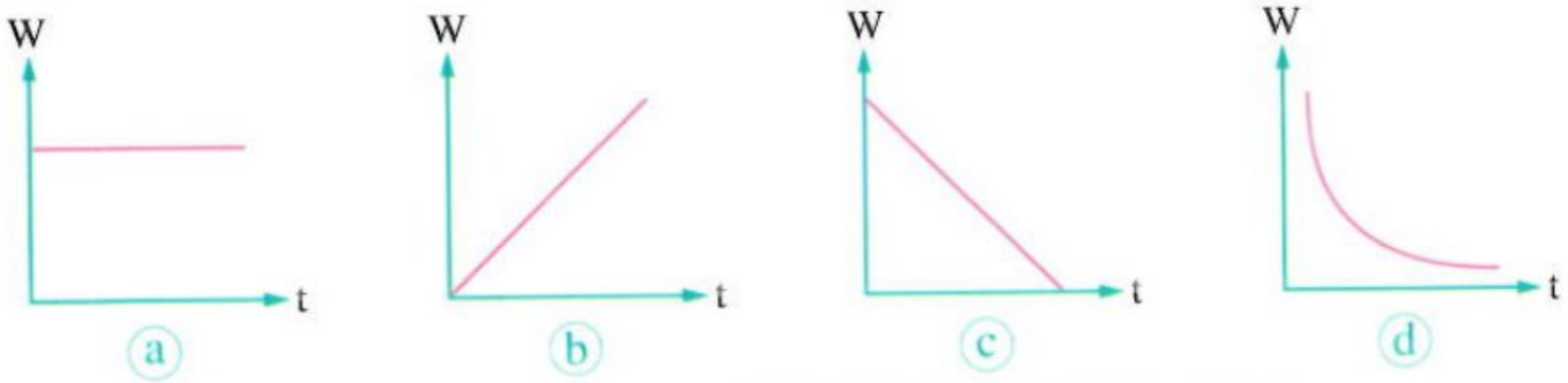


(a) 2 Ω
(b) 5 Ω
(c) 50 Ω
(d) 0.5 Ω

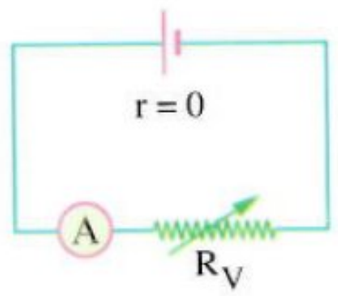
43 Which of the following graphs represents the relation between the consumed power (P_w) in a conductor that carries a direct current (DC) and time (t)?

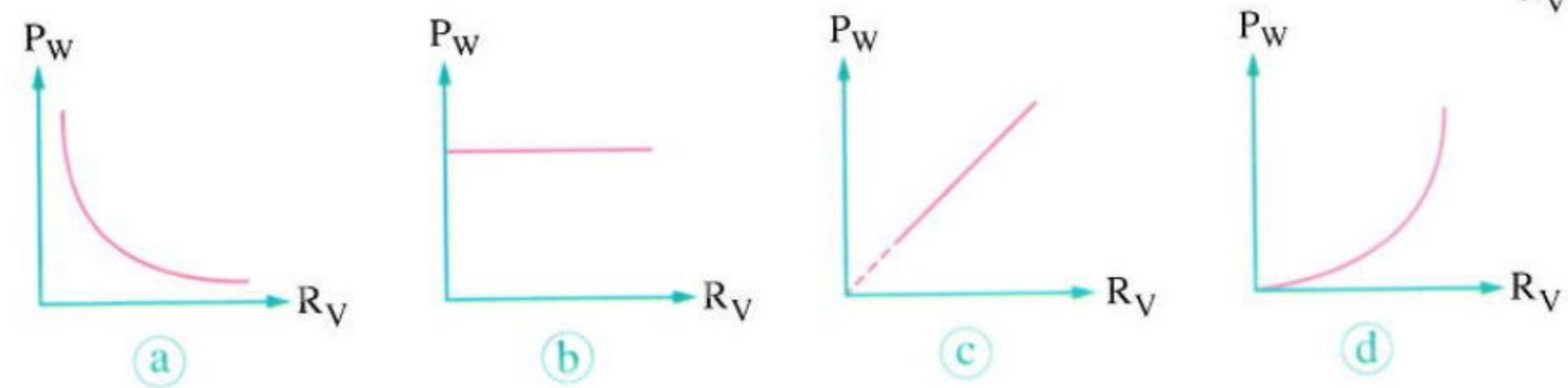


44 Which of the following graphs represents the relation between the consumed energy (W) in a conductor that carries a direct current (DC) and time (t)?



45 The opposite figure represents an electric circuit, which of the following graphs represents the relation between the consumed power through resistance R_V and its value?





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The electric resistance

46 A metal wire of length 20 m and cross-sectional area 0.5 cm^2 has a material resistivity of 10^{-7} Ω.m, so the electrical resistance of the wire equals ..... .

(a) 0.02 Ω (b) 0.04 Ω (c) 0.06 Ω (d) 0.08 Ω

47 A wire of iron which is 3.14 m long and has a radius of 0.5 mm is connected between the terminals of a 5 V battery. If the resistivity of the iron is 10^{-7} Ω.m, so the passing electric current intensity in the wire equals ..... . (Knowing that: π = 3.14)

(a) 6.2 A (b) 8.2 A (c) 9.6 A (d) 12.5 A

48 * A wire of length 200 m has a resistivity of 3.14×10^{-7} Ω.m, if the wire allows the flow of 2×10^{19} electrons per second when it is connected to a source of 64 V, the radius of the wire equals ..... . (Knowing that: π = 3.14)

(a) 10^{-1} m (b) 10^{-2} m (c) 10^{-3} m (d) 10^{-4} m

49 If the length of a copper wire is doubled and its cross-sectional area is decreased to its half, then its resistance ..... .

(a) doubles (b) decreases to its half

(c) quadruples (d) decreases to its quarter

50 When the length of a conductor has doubled and its cross-sectional area has decreased to half its original value, then the resistivity of its material ..... .

(a) is quadrupled (b) is tripled

(c) gets halved (d) remains unchanged

51 Three identical wires x, y, and z, each is connected between a potential difference (V), while being kept at temperatures 3t, 2t, and t respectively for an interval of time. In which of these wires:

(i) free electrons are colliding with the wire atoms at a higher rate?

(a) x (b) y

(c) z (d) is equal in the three wires

(ii) a higher current is flowing?

(a) x (b) y

(c) z (d) is equal in the three wires

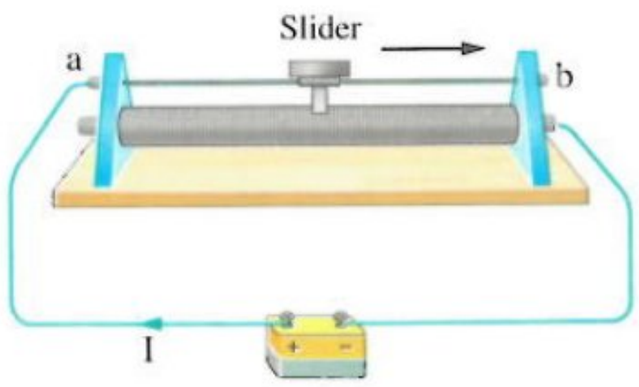
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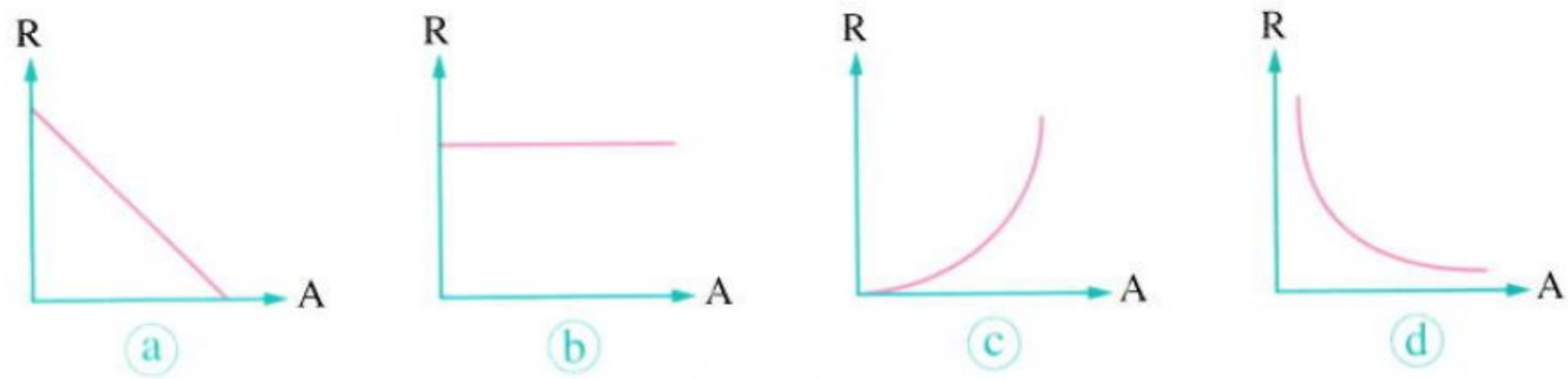
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52 In the following electric circuit, when the position of the rheostat slider is changed from its position towards position b, which of the following choices represents what happens in the circuit?

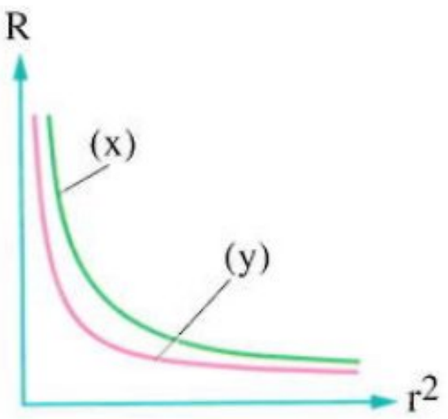
	The length of the rheostat wire that carries an electric current	The intensity of the current that passes in the circuit
a	Increases	Increases
b	Increases	Decreases
c	Decreases	Increases
d	Decreases	Decreases



53 Which of the following graphs represents the relation of resistance (R) for several copper wires that have the same length versus the cross-sectional area (A) of each?



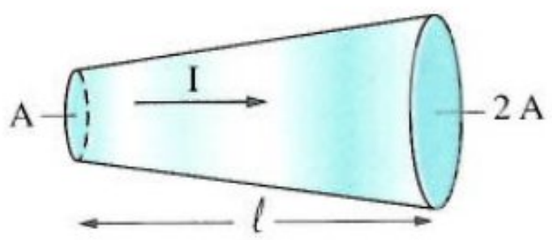
54 A group of wires, all of the same length, are made of either metal (x) or metal (y). The opposite graph represents the relationship between electrical resistance (R) and the square of the cross-sectional radius (r^2) for each wire at the same temperature. Which of the following relationships is correct when comparing the conductivity (σ) of both metals?



a $\sigma_x > \sigma_y$ b $\sigma_x < \sigma_y$

c $\sigma_x = \sigma_y$ d The answer can't be determined.

55 The opposite diagram shows a conductor whose resistivity is ρ_e. If the cross-sectional areas of its two terminals are A and 2 A, the value of its resistance is ..... .



a equal to $\frac{\rho_e \ell}{A}$ b greater than $\frac{\rho_e \ell}{A}$ c less than $\frac{\rho_e \ell}{A}$ d equal to $\frac{\rho_e \ell}{2A}$

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56 The resistivity of copper = 1.8×10^{-8} Ω.m, which means that ..... .

(a) the resistance of a copper wire that is 1 m long and has a 1 m diameter equals 1.8×10^{-8} Ω

(b) the resistance of a copper wire that is 1 m long and has a 1 m^2 cross-sectional area equals 1.8×10^{-8} Ω

(c) the resistance of a unit length of copper equals 1.8×10^{-8} Ω/m

(d) the resistance of a unit volume of copper equals 1.8×10^{-8} Ω/m^3

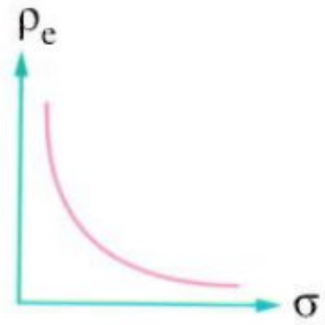
57 If the resistivity of copper is 1.8×10^{-8} Ω.m, then which of the following wires represents a copper wire of a cross-sectional area 10 mm^2?

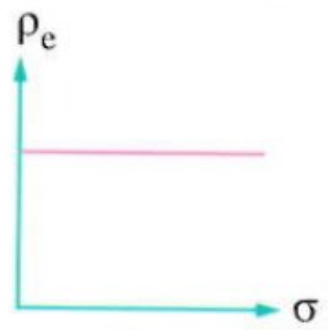
	Its length	Its resistance
(a)	10 m	1.8×10^{-8} Ω
(b)	10 m	0.018 Ω
(c)	1 m	1.8×10^{-4} Ω
(d)	1 m	1.8 Ω

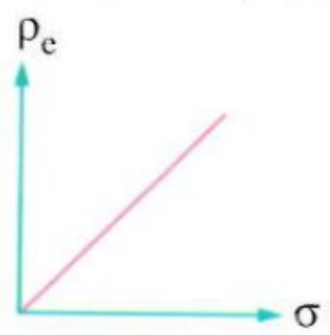
58 ✱ Two wires are of the same length and the same resistance but one of them is made of copper and the other is made of iron, so the ratio between their radii $\left(\frac{r_{Fe}}{r_{Cu}}\right)$ equals ..... .

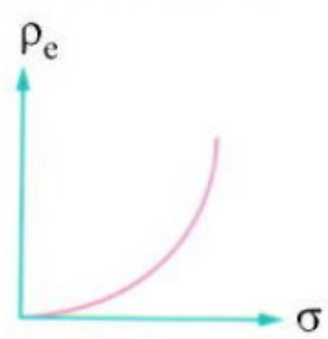
(a) $\frac{(\rho_e)_{Fe}}{(\rho_e)_{Cu}}$ (b) $\frac{(\rho_e)_{Fe}}{\sqrt{(\rho_e)_{Cu}}}$ (c) $\frac{\sqrt{(\rho_e)_{Fe}}}{(\rho_e)_{Cu}}$ (d) $\frac{\sqrt{(\rho_e)_{Fe}}}{\sqrt{(\rho_e)_{Cu}}}$

59 Which of the following graphs represents the variation of resistivity (ρ_e) for a group of wires of different materials versus the electric conductivity (σ) for each of them?

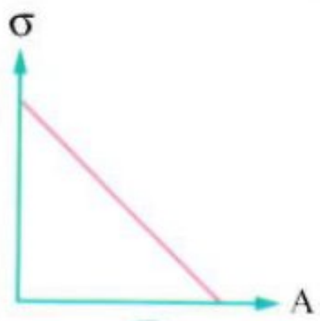
(a) 

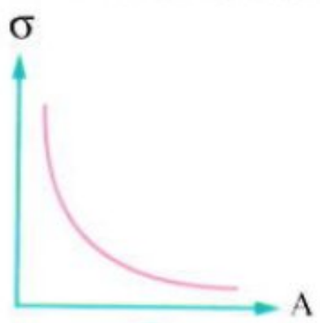
(b) 

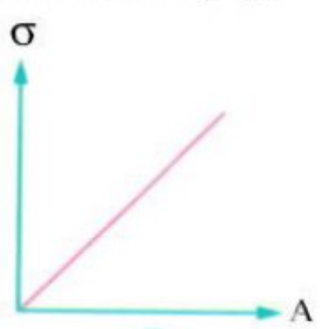
(c) 

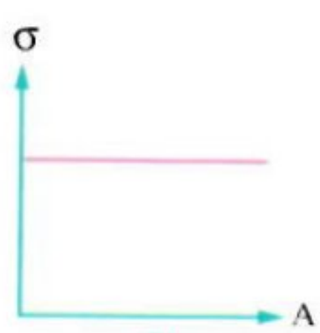
(d) 

60 Which of the following figures represents the relation between the conductivity (σ) of the material of a conductor and its cross-sectional area (A)?

(a) 

(b) 

(c) 

(d) 

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61 ✱ A wire is 106.3 cm long whose cross-sectional area is 1 mm^2 and its resistance is 1 Ω, so:

(i) The resistivity of the wire's material equals ..... .

(a) 9.41×10^{-7} Ω.m (b) 8.53×10^{-7} Ω.m (c) 5.71×10^{-6} Ω.m (d) 6.25×10^{-6} Ω.m

(ii) The conductivity of the wire's material equals ..... .

(a) $1.89 \times 10^{6}\ \Omega^{-1}.m^{-1}$ (b) $2.35 \times 10^{9}\ \Omega^{-1}.m^{-1}$

(c) $7.35 \times 10^{8}\ \Omega^{-1}.m^{-1}$ (d) $1.06 \times 10^{6}\ \Omega^{-1}.m^{-1}$

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62 ✱ The following figures represent four metallic conductors made of different materials that have the same cross-sectional area. Below each figure, the conductivity of the material of each conductor is written. Which of these conductors has the largest electrical resistance?



The conductivity of the conductor material = σ	The conductivity of the conductor material = 2 σ	The conductivity of the conductor material = 2 σ	The conductivity of the conductor material = $\frac{\sigma}{2}$
(a)	(b)	(c)	(d)

---

63 ✱ A power station is connected to a factory 2.5 km away by two wires. If the potential difference between the terminals of the two wires at the station is 240 V and that between the other terminals of the two wires at the factory is 220 V and the factory is using a current of 80 A, so: (Knowing that: π = 3.14)

(i) The resistance of one meter of the wire equals ..... .

(a) 5×10^{-5} Ω/m (b) 6×10^{-5} Ω/m (c) 12×10^{-5} Ω/m (d) 1×10^{-4} Ω/m

(ii) The radius of the wire if the resistivity of its material is 1.57×10^{-8} Ω.m equals ..... .

(a) 0.004 m (b) 0.007 m (c) 0.01 m (d) 0.05 m

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64 If you have four wires of the same material but different in both length and thickness, each of them is connected separately with the same battery and ammeter, then the ammeter reading becomes the highest in the case of the ..... wire.

(a) longest and thickest (b) longest and thinnest

(c) shortest and thickest (d) shortest and thinnest

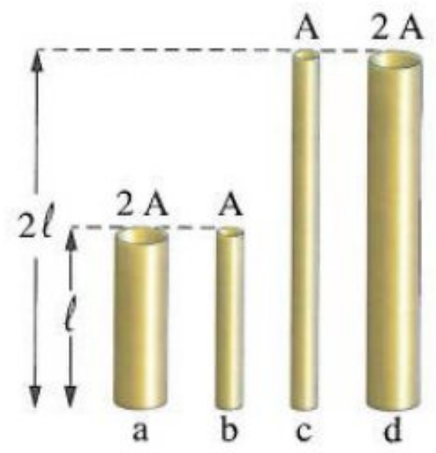
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65 A metal wire of uniform cross-section has a length ℓ. When the potential difference between the ends of the wire is V, an electric current of intensity I flows through it. If a part (X) of the wire is cut off, and the remaining part is connected between the same potential difference (V) so that the current flowing through it becomes $\frac{4}{3}$ I. Then, the length of the part that was cut (X) equals ..... .

(a) $\frac{1}{4}\ell$ (b) $\frac{1}{3}\ell$ (c) $\frac{1}{2}\ell$ (d) $\frac{2}{3}\ell$

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66 The opposite figure represents the lengths and cross-sectional areas of four wires which are made of the same material. So, at the same temperature if each of them is connected between the same electric potential difference, the correct order of the wires according to the passing electric current intensity in each of them is .....

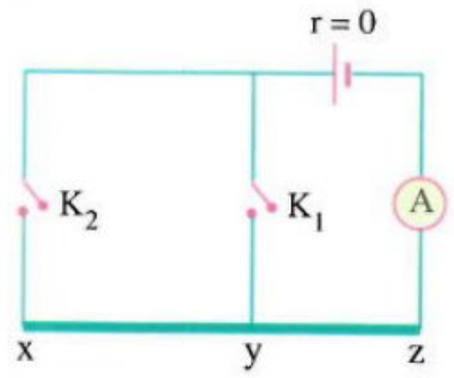


(a) c > b = d > a (b) a > b = d > c (c) b > a = c > d (d) d > a = c > b

67 * The following table shows the specifications and temperatures of four conductors x, y, z and k of uniform cross-sections made of the same metal. Which of these conductors has the highest electrical resistance?

	Conductor	Conductor length (m)	Conductor cross-sectional area (mm^2)	Conductor temperature (°C)
(a)	x	2	4	20°
(b)	y	2	4	50°
(c)	z	4	2	20°
(d)	k	4	2	50°

68 A wire xyz of uniform cross-section and electrical resistance (R) is connected in an electrical circuit as shown in the opposite figure. When only switch (K_1) is closed, the ammeter reading is 4 A. And when only switch (K_2) is closed, the ammeter reading is 1 A. so the ratio between the lengths of the two parts of the wire $\left(\frac{\ell_{xy}}{\ell_{yz}}\right)$ is equal to ..... .



(a) 2 (b) 3 (c) 4 (d) 5

69 If three copper wires z, y and x have lengths of 1 m, 4 m and 2 m respectively while having the same cross-sectional area, therefore the ratio of resistances of the three wires respectively are .....

(a) 1 : 4 : 2 (b) 4 : 1 : 2 (c) 2 : 4 : 1 (d) 2 : 1 : 4

70 * Two wires (x) and (y) are made of the same metal and with the same length. The radii of their cross-sections are r_x and r_y. Each of the wires is wound into a circular coil with radii r_1 and r_2, respectively. The ratio between the electrical resistances of the two wires $\left(\frac{R_x}{R_y}\right)$ is equal to ..... .

(a) $\frac{r_y}{r_x}$ (b) $\frac{r_1}{r_2}$ (c) $(\frac{r_y}{r_x})^2$ (d) $(\frac{r_1}{r_2})^2$

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71 ✳ Two wires are of the same material, the length of the second wire equals double the length of the first wire and its diameter equals the radius of the first wire, so the ratio between the resistance of the second wire and that of the first wire equals ..... .

(a) $\frac{1}{8}$ (b) $\frac{8}{1}$ (c) $\frac{1}{2}$ (d) 2

72 A conductor of uniform cross-section whose length is 20 m has a resistance of 108 Ω, so if another conductor of the same material has a length of 5 m while its cross-sectional area is three times as large as that of the first conductor, then the resistance of the second conductor equals ..... .

(a) 81 Ω (b) 27 Ω (c) 9 Ω (d) 6 Ω

73 ✳ A metallic rod in the shape of a cylinder of cross-sectional area 2 cm^2 and resistance 22.5 Ω is drawn uniformly until its cross-sectional area became 1.5 cm^2, so its resistance will be ..... .

(a) 37 Ω (b) 40 Ω (c) 52 Ω (d) 56 Ω

74 ✳ If a metallic wire was drawn uniformly until its length has increased to the double, then its resistance gets ..... compared to its original value.

(a) doubled (b) halved (c) quadrupled (d) quartered

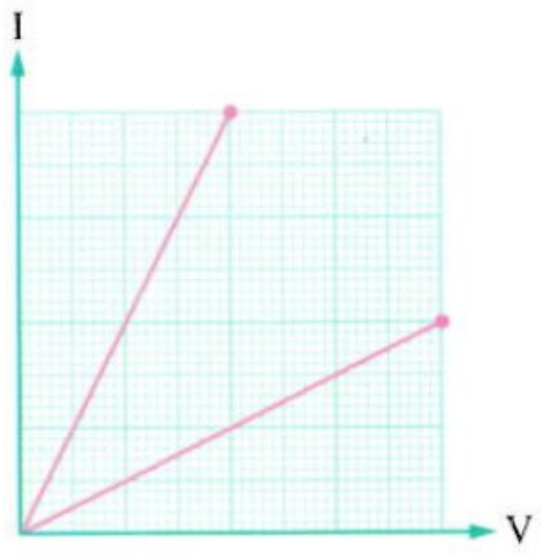
75 A metal wire is stretched uniformly such that its radius decreases by 20%, so the electrical resistance of the wire will approximately increase by ..... .

(a) 44% (b) 56% (c) 144% (d) 244%

76 A cylindrical metallic conductor with a cross-sectional diameter d and an electrical resistance R was uniformly stretched till it became a cylindrical conductor with a cross-sectional diameter $\frac{d}{x}$, so the electrical resistance of the conductor becomes ..... .

(a) x^2R (b) x^4R (c) $\frac{R}{x^2}$ (d) $\frac{R}{x^4}$

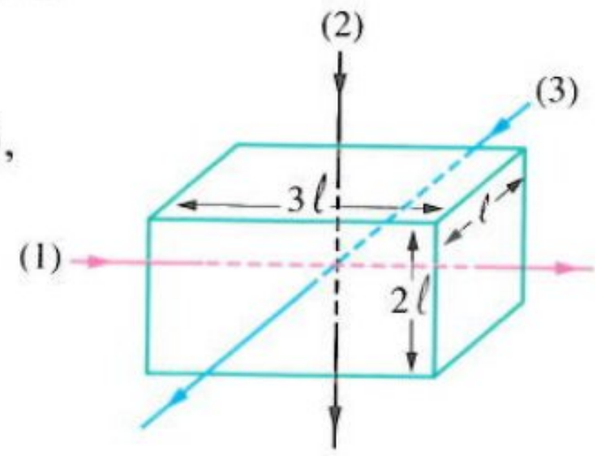
77 ✳ A wire of uniform cross-section was uniformly drawn, and the relation between the current intensity (I) through it and the potential difference (V) across its ends before and after it has been drawn is represented as in the opposite graph. Then, the wire was drawn such that ..... .



(a) its length increased to double
(b) its length increased to nine times
(c) its cross-sectional area decreased to one-third
(d) its cross-sectional area decreased to one-ninth

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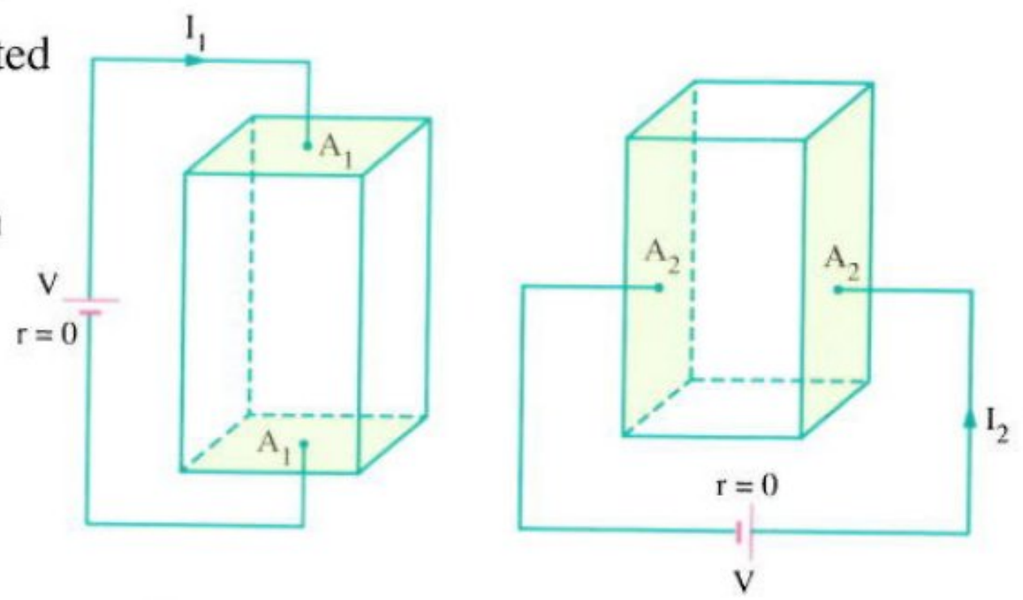
78 ✱ The opposite figure shows a conducting solid cuboid where any pair of opposite faces of the cuboid can be connected between the terminals of an electric source and the paths (1), (2) and (3) are the probable pathways of electric currents through the conductor, in which of these paths the electric current faces the highest resistance?



(a) Path (1) (b) Path (2)

(c) Path (3) (d) All paths have the same resistance.

79 ✱ An electrical conductor in the shape of a solid rectangular cuboid is connected in two different ways to a battery of negligible internal resistance, as shown in circuits (1) and (2). Then, the ratio between the current intensities in the two circuits $\left(\frac{I_1}{I_2}\right)$ is equal to ..... .



Circuit (1) Circuit (2)

(a) $\frac{A_1}{A_2}$ (b) $\left(\frac{A_1}{A_2}\right)^2$

(c) $\frac{A_2}{A_1}$ (d) $\left(\frac{A_2}{A_1}\right)^2$

80 ✱ A metal wire of volume 2×10^{-4} m^3 and cross-sectional area 4×10^{-5} m^2 has a resistance of 1.25 Ω, so its conductivity equals ..... .

(a) $10^3\ \Omega^{-1}.m^{-1}$ (b) $10^5\ \Omega^{-1}.m^{-1}$

(c) $10^7\ \Omega^{-1}.m^{-1}$ (d) $10^8\ \Omega^{-1}.m^{-1}$

81 A metallic cube has a side length ℓ, when it is connected to an electric circuit at two opposite faces, its electrical resistance equals R, so the electrical conductivity (σ) of this metal can be determined using the relation; .....

(a) $\sigma = R\ell$ (b) $\sigma = \frac{\pi}{R\ell}$ (c) $\sigma = \frac{1}{2R\ell}$ (d) $\sigma = \frac{1}{R\ell}$

82 ✱ Two wires are made of copper, the length of the first wire is 10 m and its mass is 0.1 kg, while length and mass of the second wire are 40 m and 0.2 kg, respectively, so the ratio between their resistances $\left(\frac{R_1}{R_2}\right)$ equals ..... .

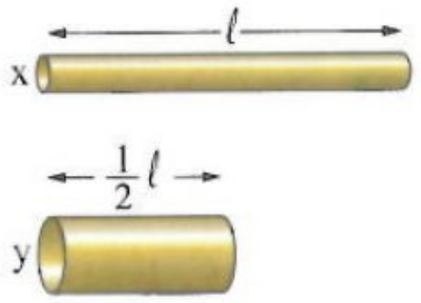
(a) $\frac{1}{4}$ (b) $\frac{1}{5}$ (c) $\frac{1}{7}$ (d) $\frac{1}{8}$

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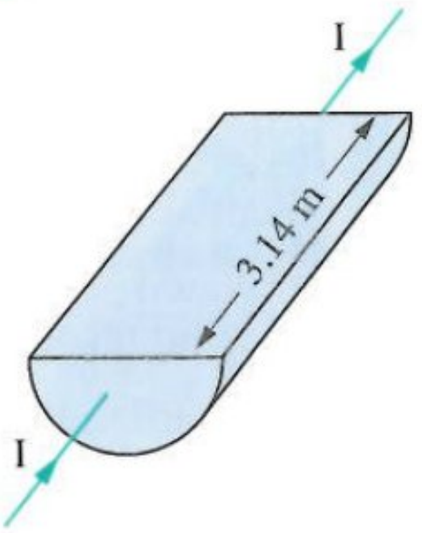
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83 ✱ The opposite figure shows two wires x and y of the same material. If the mass of wire x is 5 g and the mass of wire y is 10 g, then the ratio between the resistances of the two wires $\left(\frac{R_x}{R_y}\right)$ equals ..... .



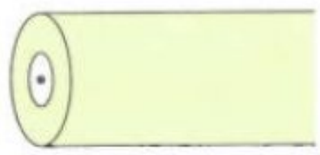
(a) $\frac{2}{1}$ (b) $\frac{4}{1}$ (c) $\frac{8}{1}$ (d) $\frac{16}{1}$

84 ✱ In the opposite figure, a conductor that has a length of 3.14 m and its cross-section is in the shape of a semi-circle with a diameter of 2 mm carries an electric current. If you know that the resistivity of the conductor material is 10^{-6} Ω.m, then its electrical resistance equals ..... . (Given that: $\pi = 3.14$)



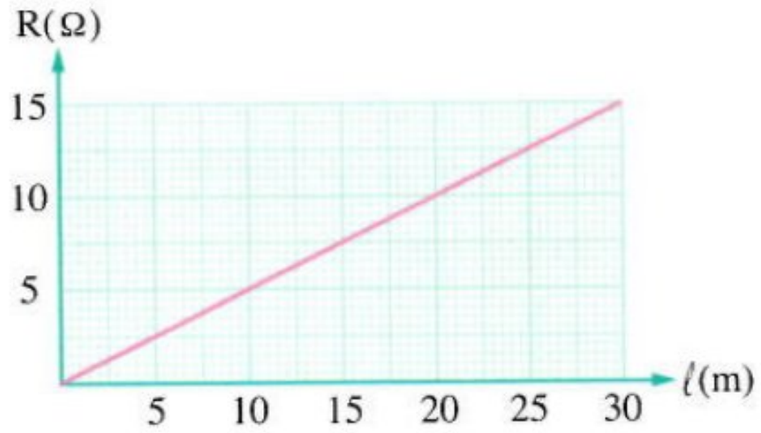
(a) 0.25 Ω (b) 0.5 Ω
(c) 1 Ω (d) 2 Ω

85 A hollow cylinder made of a metal with resistivity 2×10^{-8} Ω.m has a length of 1 m and inner and outer radii of 1 mm and 2 mm respectively, as shown in the opposite figure, so the electrical resistance between the ends of the cylinder is equal to ..... .



(a) 1.6×10^{-3} Ω (b) 2.1×10^{-3} Ω (c) 4.8×10^{-3} Ω (d) 8×10^{-3} Ω

86 ✱ The opposite graph represents the relation between the resistance (R) of a wire and its length (ℓ). If the cross-sectional area of the wire is 0.1 cm^2, so:



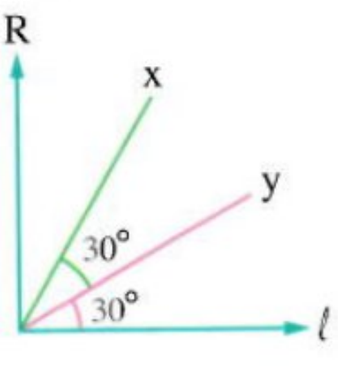
(i) The resistivity of the wire's material (ρ_e) equals ..... .

(a) 3×10^{-7} Ω.m (b) 5×10^{-6} Ω.m
(c) 4×10^{-5} Ω.m (d) 9×10^{-8} Ω.m

(ii) The resistance of a 25 m long section of this wire equals ..... .

(a) 9.25 Ω (b) 11.3 Ω (c) 12.5 Ω (d) 15.9 Ω

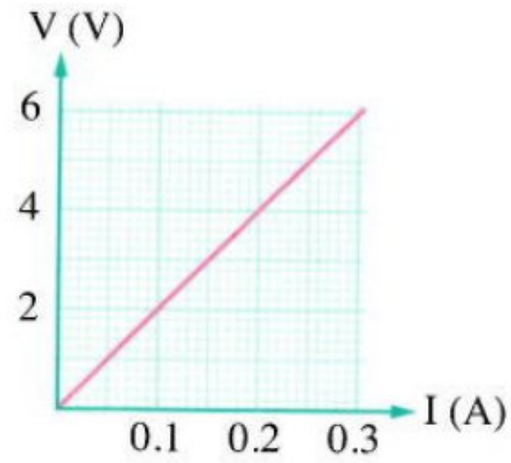
87 The opposite graph represents the variation of resistance (R) with length (ℓ) for two copper wires (x) and (y), so the ratio between the cross-sectional areas of the two wires $\left(\frac{A_x}{A_y}\right)$ is ..... .



(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{3}{1}$ (d) $\frac{\sqrt{3}}{1}$

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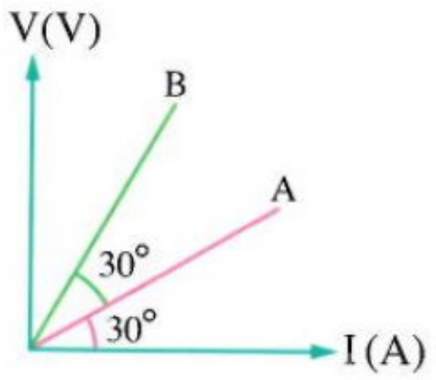
88 ✱ The opposite graph represents the variation of the potential difference (V) across a wire of length 5 m and cross-sectional area 0.1 mm^2 versus the current intensity (I) through the wire, so the conductivity of the wire's material equals ..... .



(a) $4.1 \times 10^6\ \Omega^{-1}.m^{-1}$ (b) $3.2 \times 10^5\ \Omega^{-1}.m^{-1}$

(c) $2.5 \times 10^6\ \Omega^{-1}.m^{-1}$ (d) $1.6 \times 10^5\ \Omega^{-1}.m^{-1}$

89 ✱ The opposite graph shows the variation of the electric potential difference (V) with the electric current intensity (I) in two wires A and B of the same material, so:



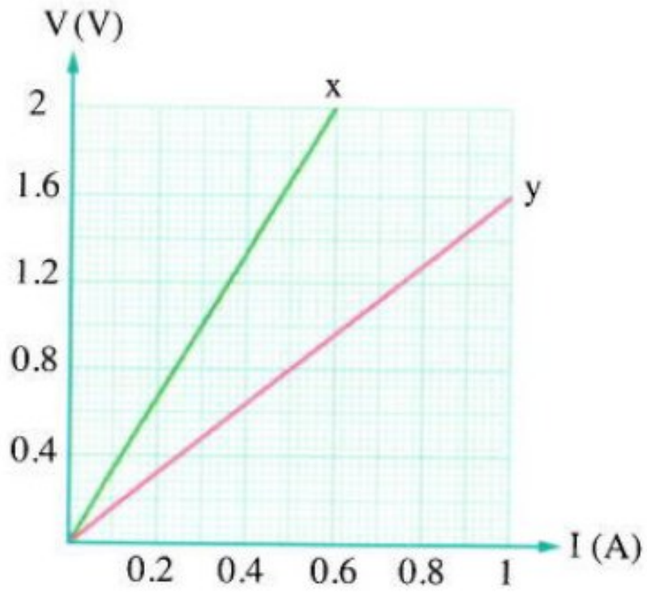
(i) If both wires have the same length and the cross-sectional area of wire B is $3 \times 10^{-6}\ m^2$, the cross-sectional area of wire A equals ..... .

(a) $10^{-6}\ m^2$ (b) $9 \times 10^{-6}\ m^2$ (c) $3 \times 10^{-6}\ m^2$ (d) $12 \times 10^{-8}\ m^2$

(ii) If both wires have the same cross-sectional area and the length of wire B is 3 m, the length of wire A equals ..... .

(a) 1 m (b) 2 m (c) 3 m (d) 9 m

90 ✱ The opposite graph represents the variation of the potential difference (V) across two wires (x and y) that have the same length with the current intensity (I) in both of them at constant temperature.



If the ratio between the cross-sectional areas of the two wires $\left(\frac{A_x}{A_y}\right)$ equals $\frac{12}{25}$, so the ratio between the resistivities of their materials $\left(\frac{(\rho_e)_x}{(\rho_e)_y}\right)$ equals ..... .

(a) $\frac{2}{1}$ (b) $\frac{1}{1}$ (c) $\frac{1}{4}$ (d) $\frac{1}{7}$

91 The opposite table shows different values of lengths, cross-sectional areas and resistivities for wires that are made of different materials:

	Wire length ℓ (m)	Cross-sectional area A (cm^2)	Resistivity $\rho_e \times 10^{-4}$ (Ω.m)
(1)	10	0.1	0.05
(2)	5	0.5	0.25
(3)	5	0.1	0.5
(4)	0.5	0.5	0.005

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(i) Which of these wires carries a current of intensity 2 A when the potential difference between its terminals equals 10 V?

(a) Wire (1) (b) Wire (2) (c) Wire (3) (d) Wire (4)

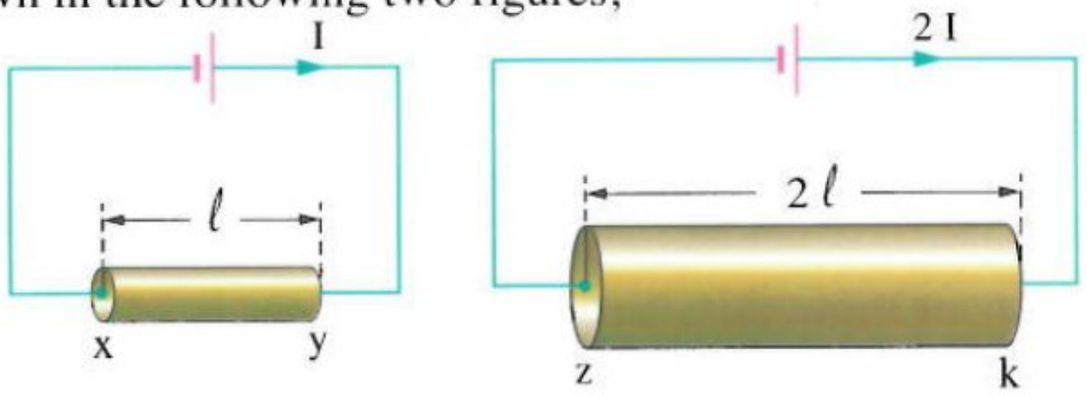
(ii) Which of these wires gives a larger amount of heat than the other wires when the same current intensity passes through them for the same time interval?

(a) Wire (1) (b) Wire (2) (c) Wire (3) (d) Wire (4)

(iii) Which of these wires gives less thermal power than the other wires when each of them is connected between the same potential difference?

(a) Wire (1) (b) Wire (2) (c) Wire (3) (d) Wire (4)

92 Each of conductors xy and zk has a uniform cross-section and both of them are made of the same metal with lengths ℓ and 2ℓ respectively. Each is connected in an electrical circuit as shown in the following two figures,



If the electrical powers consumed in the two conductors are equal, the ratio between the cross-sectional areas of the conductors $\left(\frac{A_{xy}}{A_{zk}}\right)$ equals ..... .

(a) $\frac{1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{1}{6}$ (d) $\frac{1}{8}$

93 A wire of resistance R consumes an electric power P_w when the potential difference between its terminals is V, if the wire is drawn uniformly so that its length gets doubled while keeping the potential difference between its terminals V, it will consume an electric power of ..... .

(a) $2\,P_w$ (b) $4\,P_w$ (c) $\frac{P_w}{2}$ (d) $\frac{P_w}{4}$

94 * A wire of length 2 m has a cross-sectional area of 4×10^{-6} m^2, when the potential difference between its terminals was 20 V, the consumed power through it was 10 W, so:

(i) The resistivity of the wire's material equals ..... .

(a) 2×10^{-7} Ω.m (b) 10^{-5} Ω.m

(c) 8×10^{-5} Ω.m (d) 4×10^{-6} Ω.m

(ii) The number of electrons that pass through a cross-section of the wire in one minute equals ..... . (Where: $e = 1.6 \times 10^{-19}$ C)

(a) 9.741×10^{17} electrons (b) 6.435×10^{18} electrons

(c) 2.314×10^{19} electrons (d) 1.875×10^{20} electrons

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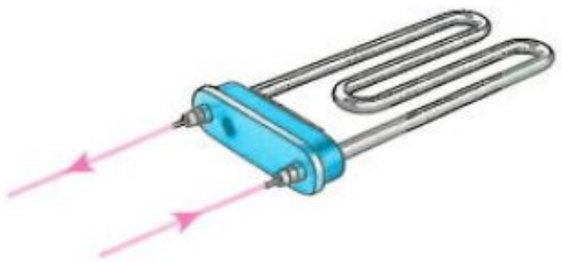
Second Essay questions

Electric current intensity

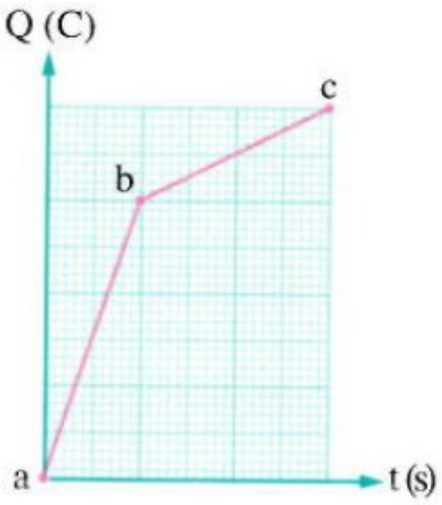
1. **Explain why** some solid materials conduct the electric current while others are electrically insulators.

2. A current of intensity 3.2 mA passes through a wire, **calculate** the rate of electrons flow across a section of the wire.

3. The opposite figure shows a wire of a heater through which a direct electric current is passing. **Explain why** the intensity of the electric current at the ends of the wire is the same, although electric energy is consumed through it.

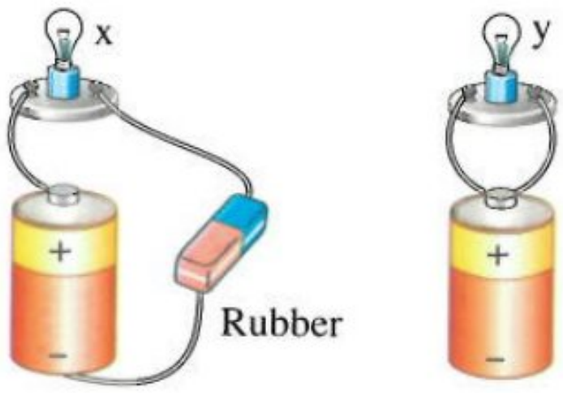


4. The opposite graph represents the relation between the quantity of electric charges (Q) passing across a certain section of a conductor and time (t), **deduce the ratio between** intensities of the current passing in the conductor during the intervals ab, bc respectively.



Electric potential difference

5. **Explain:** None of the two lamps x and y is lit up, even though each of them is connected to a dry cell as shown in the opposite figure.



6. In the opposite figure, if the work done to transfer 10^{20} free electrons from point y to point x is 80 J, **calculate** the electric potential at point y.



Ohm's law

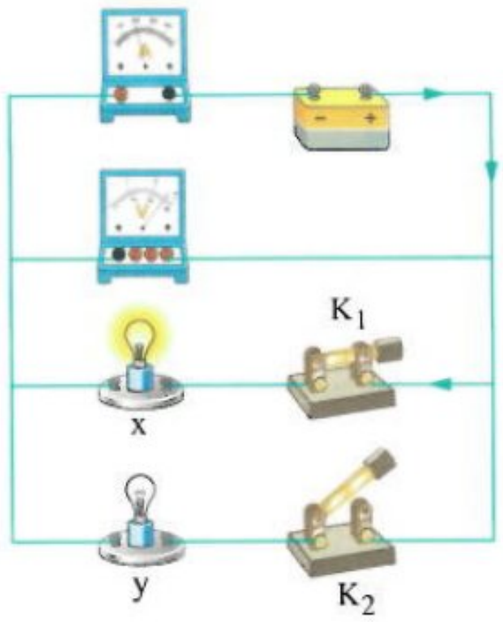
7. The potential difference between the terminals of a battery is 9 V when it is connected to an electric lamp whose filament resistance is 1.6 Ω, **calculate** the number of electrons passing per minute across a section of the filament.

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8 In the opposite circuit, an electric battery of negligible internal resistance is connected to two lamps x and y through two switches K_1 and K_2 respectively, with an ammeter and a voltmeter. If the resistance of lamp y is more than that of lamp x, **illustrate what happens to** the reading of each of the ammeter and voltmeter when switch K_1 is opened and switch K_2 is closed.



Electric power and electric energy

9 Two metal wires, the first has a resistance R where 10^{20} electrons pass across a section of it per second and the second has a resistance 2 R where 2×10^{20} electrons pass across a section of it per second. **Calculate** the ratio between the consumed power in the first wire and the consumed power in the second wire.

10 An electric heater whose coil is connected to an electrical source of potential difference 220 V and negligible internal resistance to heat the water, **what** is the effect of replacing the heater coil by another of higher electrical resistance on the rate of heating the water? **Why?**

11 In the two figures (1) and (2), if the consumed powers in the two resistors R_1 and R_2 are equal and the ratio between the readings of the two ammeters $\left(\frac{A_1}{A_2}\right)$ equals $\frac{1}{2}$:

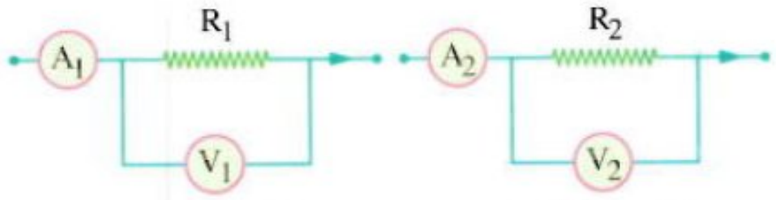


Figure (1) Figure (2)

(a) **Find** the ratio between the readings of the two voltmeters $\left(\frac{V_1}{V_2}\right)$.

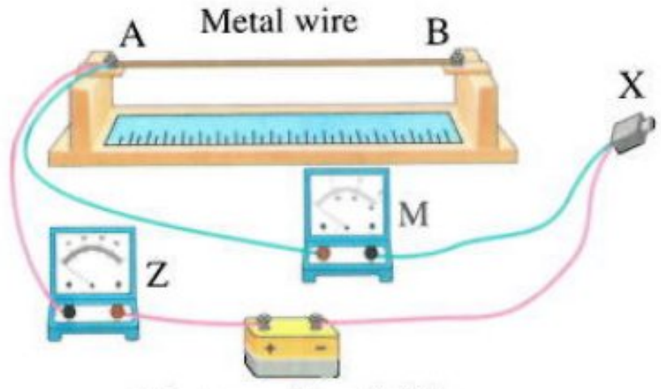
(b) **Explain why** it is not appropriate to use the readings of the two voltmeters V_1 and V_2 and the two ammeters A_1 and A_2 to satisfy Ohm's law.

The electric resistance

12 A student used the tools shown in the figure to study how the resistance of a wire (AB) of uniform cross-section changes with the change of its length:

(a) **What** is the role of terminal (X) to achieve the aim of the experiment?

(b) **What** is the aim of using the two devices Z and M?



A battery of negligible internal resistance

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13 **How** can the electric resistance of a copper wire be increased at a certain temperature?

14 **What are the physical quantities that are measured in the following units?**

(a) Ampere.second

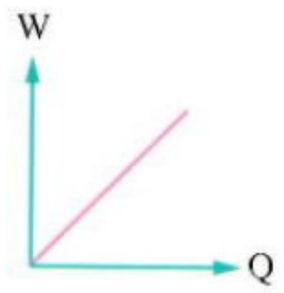
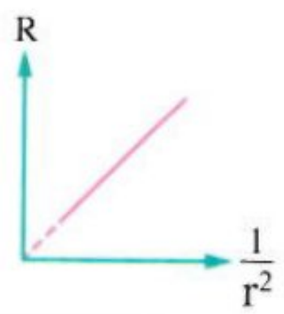
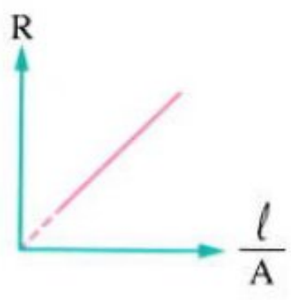
(b) Volt.coulomb

(c) Volt. ampere^{-1}

(d) Joule/ohm.coulomb

(e) Volt.second.ohm^{-1}

15 **Write the mathematical relation and the equivalent to the slope for each of the following:**

(a)	(b)	(c)
		

Where: (Q) is the quantity of electric charges, (R) is the resistance, (ℓ) is the conductor length, (A) is the conductor cross-sectional area, (r) is the conductor radius, (W) is the work.

16 The opposite table shows the specifications of three metallic conductors which are made of different materials (x, y and z) and have the same cross-sectional area if σ refers to the electric conductivity of the material:

Conductor	Conductor length	Conductor resistance
x	2 m	1 Ω
y	3 m	4 Ω
z	3 m	6 Ω

(a) **What** is the ratio between σ_x, σ_y, σ_z?

(b) **Which** one of these materials is preferred to be used in the manufacture of electric connecting wires? **Explain your answer.**

17 A solid cube of side length 10 cm made of a material whose resistivity is 10^{-7} Ω.m has been shaped into a cylindrical wire with a resistance of 20 Ω. **Calculate** the length of the wire.

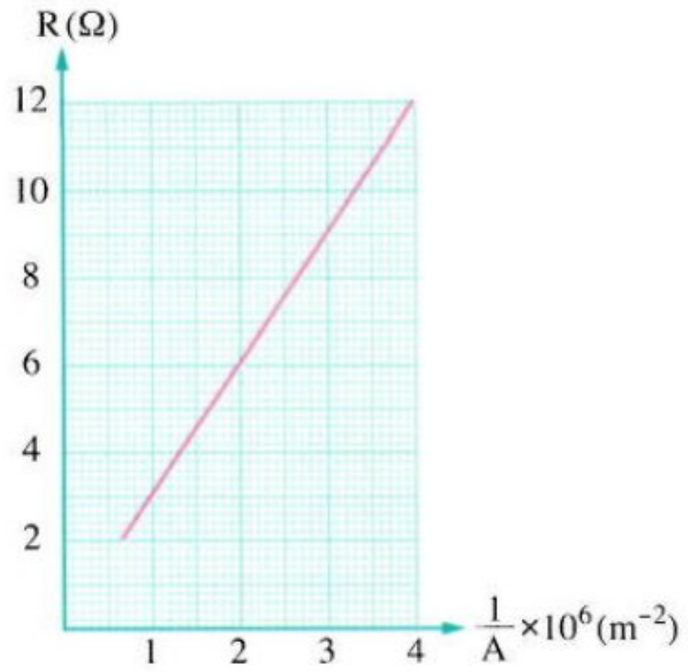
18 A wire of length 2 m has a resistance 2 Ω, if the density of the wire's material is 7000 kg/m^3 and its resistivity is 10^{-6} Ω.m. **Calculate** the mass of the wire.

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19 A group of wires made of the same material, each is 12 m long. The opposite figure represents the relation between the resistance (R) of each wire and the reciprocal of its cross-sectional area $\left(\frac{1}{A}\right)$, **find**:



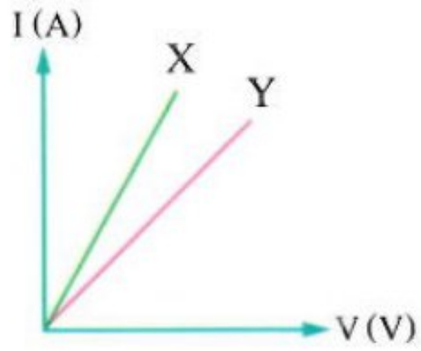
(a) The electric resistance of a wire made of the same material and having the same length as the other wires and its cross-sectional area is 0.0025 cm^2.

(b) The electric conductivity of the wires' material.

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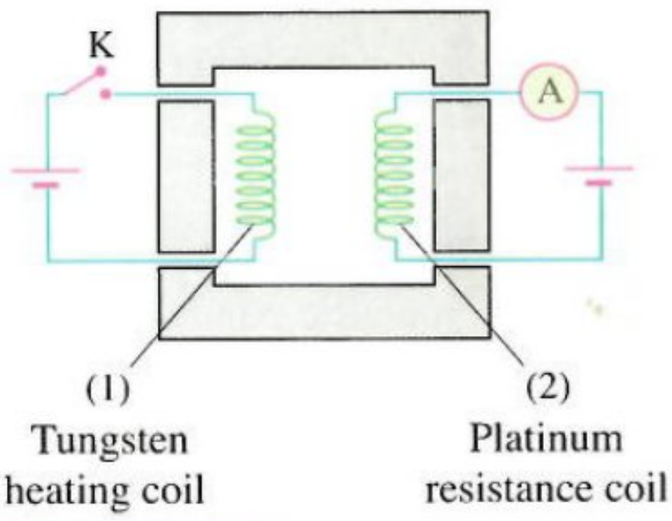
20 The opposite graph represents the relation between the electric current intensity (I) passing in two wires X, Y of uniform cross-section, made of the same material, having the same length and the potential difference (V) between the ends of each wire.

Which of them is the thickest? **And why?**



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21 In the opposite figure, a closed insulated chamber contains two coils, after closing the switch K which is connected to coil (1) for a period, it was observed that the reading of the ammeter (A) connected to the coil (2) decreased. **How can you explain that?**



---

22 An electric lamp A of power 80 W works on a potential difference of 220 V and an electric lamp B of power 20 W works on a potential difference of 24 V, given that the filaments of the two lamps are made of the same material and have the same length.

Calculate the ratio between the radii of the two filaments $\left(\frac{r_A}{r_B}\right)$.

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Questions of the Previous Years Exams on Lesson 1

Electric potential difference

1 The electric current through a metal conductor which is connected to the terminals of a voltage source is the flow of ..... . *(2nd session 25)*

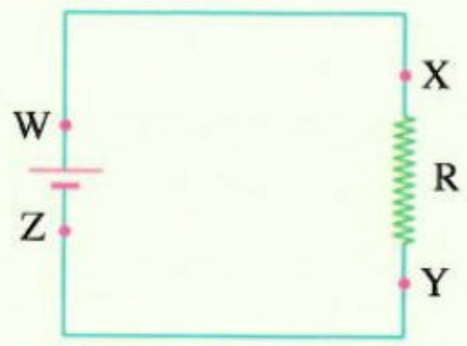
(a) free electrons from the terminal of lower electric potential to the terminal of higher potential

(b) free electrons from the terminal of higher electric potential to the terminal of lower potential

(c) positive ions from the terminal of lower electric potential to the terminal of higher potential

(d) positive ions from the terminal of higher electric potential to the terminal of lower potential

2 A simple electric circuit consisting of a battery and an ohmic resistance as shown in the figure, which of the following choices correctly expresses the relation between the electric potentials of the shown points W, X, Y, and Z?

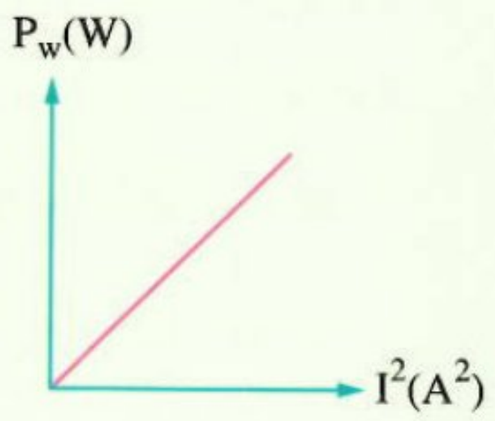


(Practice 26)

(a)	The electric potential of point W is greater than the electric potential of point Z	The electric potential of point X is less than the electric potential of point Y
(b)	The electric potential of point W is less than the electric potential of point Z	The electric potential of point X is greater than the electric potential of point Y
(c)	The electric potential of point W is greater than the electric potential of point Z	The electric potential of point X is greater than the electric potential of point Y
(d)	The electric potential of point W is less than the electric potential of point Z	The electric potential of point X is less than the electric potential of point Y

Electric power and electric energy

3 The opposite graph represents the relation between the power consumed (P_W) in a conductor and the square of the current intensity (I^2) through this conductor, so the slope of the line representing the relationship equals ..... . *(Practice 25)*



(a) the resistance of the conductor

(b) the reciprocal of the resistance of the conductor

(c) the potential difference across the conductor

(d) the square of the potential difference across the conductor

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4 An electric lamp is labelled (100 V, 80 W) which means ..... . *(Practice 25)*

(a) the electrical resistance of the lamp is 0.8 Ω

(b) the electrical resistance of the lamp is 1.25 Ω

(c) when the potential difference across the lamp is 100 V, a current of 0.8 A flows through it

(d) when the potential difference across the lamp is 100 V, a current of 1.25 A flows through it

5 If the work done to transfer a quantity of electric charge of 2 C inside and outside the electric source is 12 J, this means that ..... . *(1st session 25)*

(a) the opposition to the flow of the electric current in the source is 6 Ω

(b) the electric current intensity through the source is 6 A

(c) the electric power generated by the source is 6 W

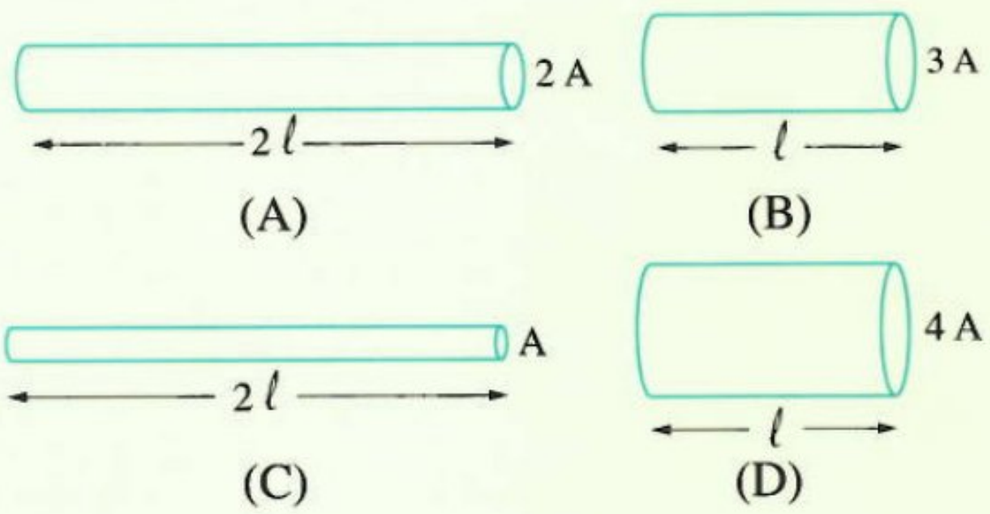
(d) the electromotive force of the source is 6 V

The electric resistance

6 Two copper wires have the same length, if the cross-sectional area of the second wire is three times that of the first wire, the ratio between the resistance of the first wire and the resistance of the second wire $\left(\frac{R_1}{R_2}\right)$ equals ..... . *(Exp. May 21)*

(a) $\frac{3}{1}$ (b) $\frac{1}{3}$ (c) $\frac{1}{6}$ (d) $\frac{6}{1}$

7 The opposite figure shows four conductors of uniform cross-sections made of the same material but with different dimensions, so their correct arrangement in an ascending order according to their electrical resistances is ..... *(Exp. May 21)*



(a) B ⟶ C ⟶ A ⟶ D

(b) D ⟶ A ⟶ C ⟶ B

(c) C ⟶ A ⟶ B ⟶ D

(d) D ⟶ B ⟶ A ⟶ C

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8 A conductor of length ℓ and cross-sectional area 3 A is connected between a potential difference V so that a current of intensity I passes through it, when the same potential difference is used with a different conductor of the same material as the first one, it was found that a current of intensity 3 I passes through it, so the length and the cross-sectional area of the second conductor are ..... .

(Exp. June 21)

	Length	Cross-sectional area
(a)	2ℓ	18 A
(b)	3ℓ	3 A
(c)	18ℓ	2 A
(d)	$\frac{1}{3}\ell$	$\frac{1}{3}$ A

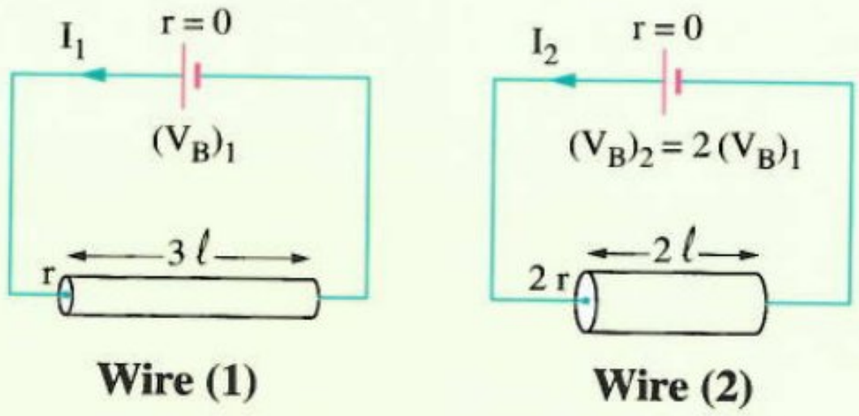
9 Two wires are made of the same material, the diameter of the first wire is 3 times the diameter of the second wire and the resistance of the second wire is 4 times the resistance of the first wire, so the length of the second wire is ..... the length of the first wire.

(a) $\frac{4}{3}$ (b) $\frac{36}{1}$ (c) $\frac{4}{9}$ (d) $\frac{12}{1}$ *(1st session 21)*

10 A current of intensity I passes through a conductor of length ℓ and cross-sectional area A, when the battery is changed with a different one, the current passing through the same conductor becomes 3 I, so the cross-sectional area of the conductor becomes ..... .

(a) A (b) 3 A (c) $\frac{1}{3}$ A (d) 6 A *(2nd session 21)*

11 Two wires (1) and (2) are made of the same material, wire (1) has length (3ℓ) and radius (r) while wire (2) has length (2ℓ) and radius (2 r) as shown in the figure: *(1st session 22)*



Wire (1) **Wire (2)**

So, the ratio $\left(\frac{I_1}{I_2}\right)$ = ...

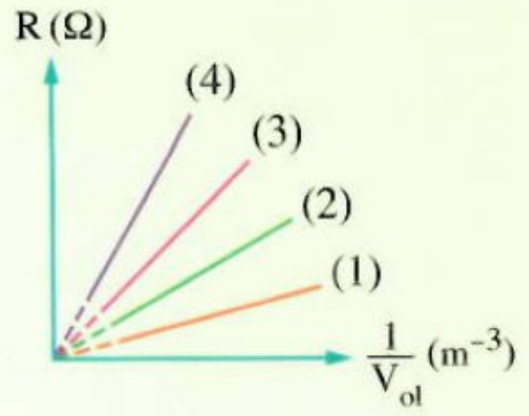
(a) $\frac{12}{1}$ (b) $\frac{1}{12}$ (c) $\frac{3}{2}$ (d) $\frac{1}{6}$

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12 The graph shows the relation between the electric resistance (R) and the reciprocal of the volume $\left(\frac{1}{V_{ol}}\right)$ for several groups of wires of the same length but of different materials. The correct arrangement of these groups of wires concerning the conductivity (σ) of their materials is ..... *(2nd session 22)*



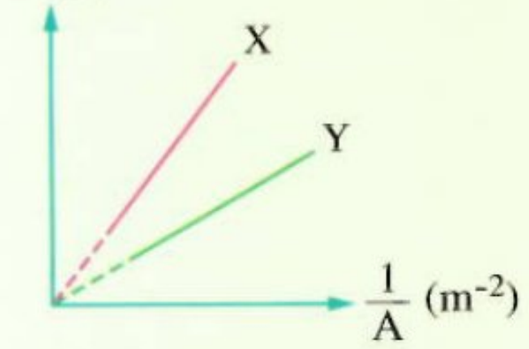
(a) $\sigma_4 > \sigma_1 > \sigma_3 > \sigma_2$

(b) $\sigma_1 > \sigma_3 > \sigma_2 > \sigma_4$

(c) $\sigma_1 > \sigma_2 > \sigma_3 > \sigma_4$

(d) $\sigma_4 > \sigma_3 > \sigma_2 > \sigma_1$

13 Two groups of wires, X and Y, are of the same length and of different metals, the opposite figure represents the relation between the electric resistance (R) of the wire and the reciprocal of the cross-sectional area $\left(\frac{1}{A}\right)$, which of the following is correct? *(Exp. 23)*



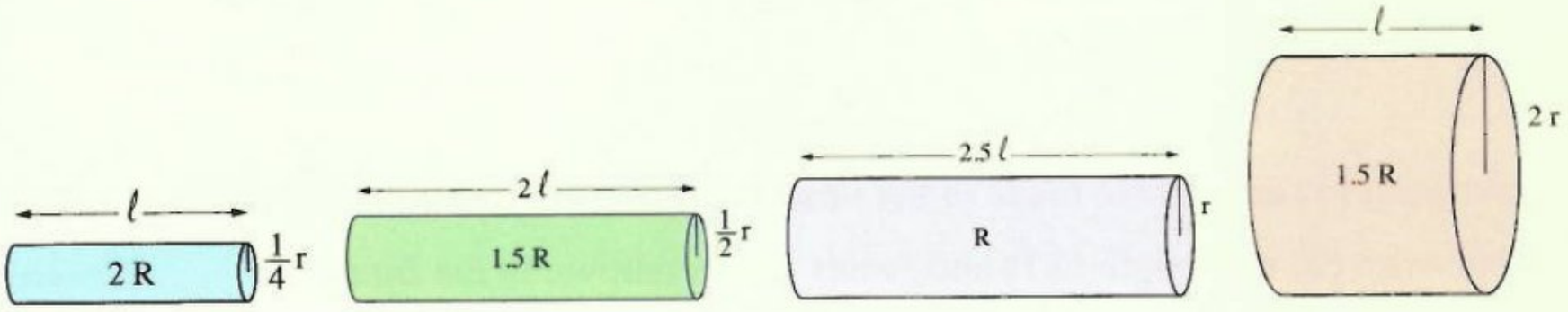
(a) Group X is made of a metal with a higher resistivity.

(b) Group Y is made of a metal with a higher resistivity.

(c) Group X is made of a metal with a higher electric conductivity.

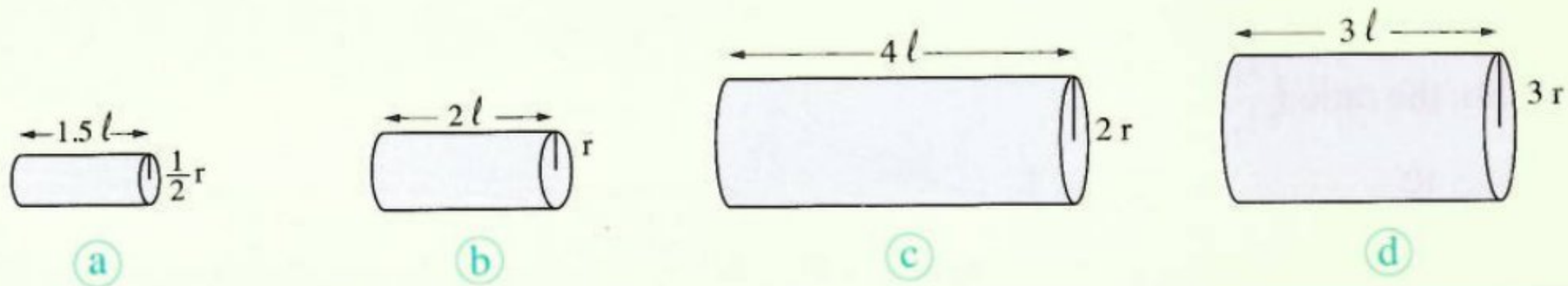
(d) The electric conductivity of the two groups is equal.

14 The following figures illustrate four wires of different materials, using the data on the figures, which wire material has the greatest conductivity at the same temperature? *(1st session 23)*



(a) (b) (c) (d)

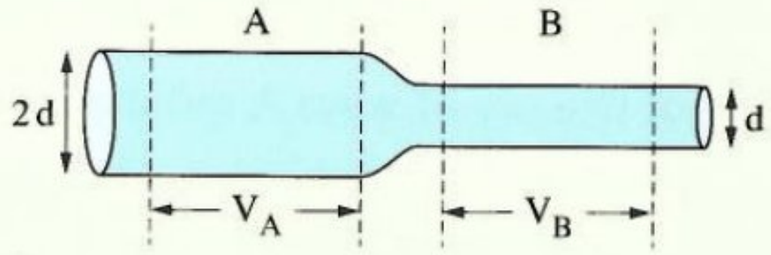
15 The following figures represent four aluminum wires. Which wire has the least electric resistance? *(2nd session 23)*



(a) (b) (c) (d)

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16 The figure represents a metallic conductor with different cross-sections A and B and electrical resistances R_A and R_B respectively. It is connected to the terminals of a battery in a closed electric circuit.

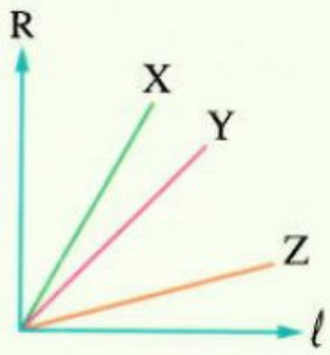


Knowing that the length of part (A) = The length of part (B).

So, the ratio: $\frac{\text{Potential difference } (V_A)}{\text{Potential difference } (V_B)}$ = ... *(1st session 24)*

(a) $\frac{R_B}{R_A}$ (b) $\frac{R_A}{R_B}$ (c) $\frac{2\,R_A}{R_B}$ (d) $\frac{4\,R_A}{R_B}$

17 The graph shows the relation between the change in the electric resistance of a number of wires made of three different materials X, Y and Z having the same cross-sectional area at the same temperature, versus length (ℓ). If σ_X, σ_Y, and σ_Z represent the electric conductivity of the wire's materials, which of the following choices is correct? *(1st session 24)*



(a) $\sigma_Z = \sigma_Y = \sigma_X$ (b) $\sigma_Z < \sigma_Y < \sigma_X$

(c) $\sigma_Z > \sigma_Y > \sigma_X$ (d) $\sigma_Z > \sigma_X > \sigma_Y$

18 Figure (1) represents an electric circuit containing a conductor AB at temperature T_1, so the voltmeter reading was V_1. When the temperature of the conductor AB is raised to T_2 and stabilizes, the reading of the voltmeter becomes V_2 as in figure (2). *(2nd session 24)*

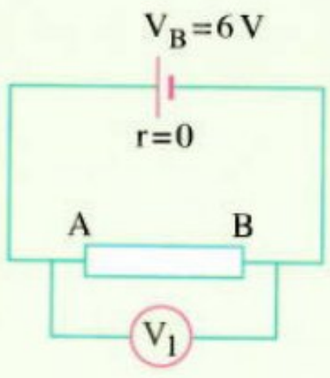


Figure (1)

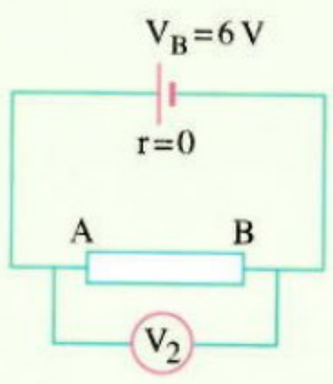


Figure (2)

Which of the following choices is correct?

(a) The reading of voltmeter V_2 = Zero

(b) The reading of voltmeter V_1 = The reading of voltmeter V_2

(c) The reading of voltmeter V_1 > The reading of voltmeter V_2

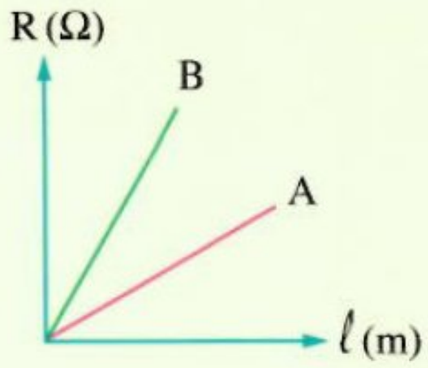
(d) The reading of voltmeter V_1 < The reading of voltmeter V_2

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19 The opposite graph shows the relation between the resistances (R) of two sets of wires A and B which are made of two different materials having the same cross-sectional area at constant temperature and their length (ℓ), so which choice is correct?



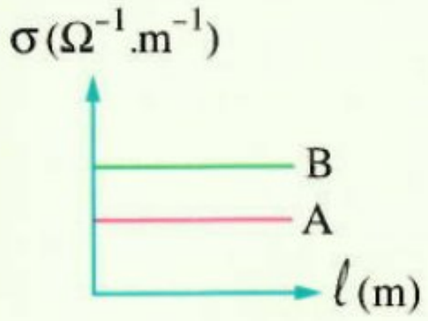


Figure (1)

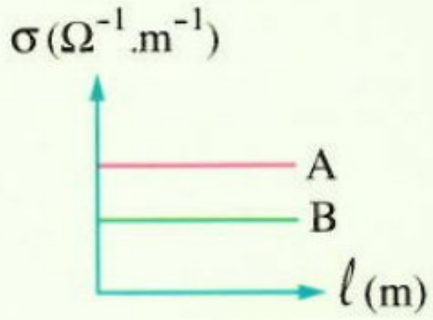


Figure (2)

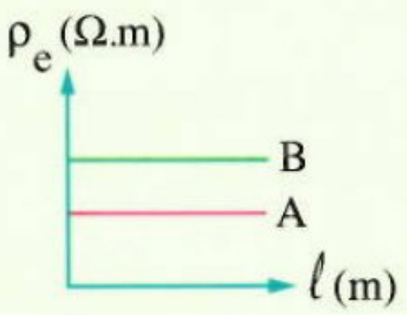


Figure (3)

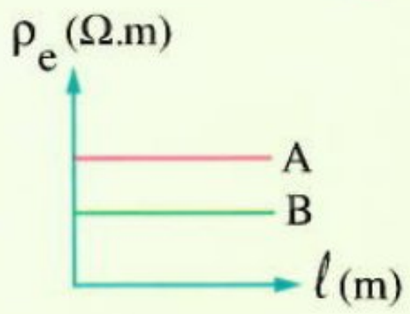
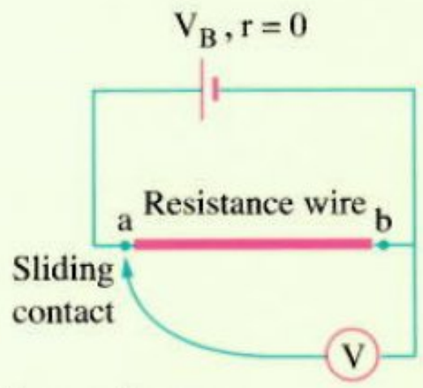


Figure (4)

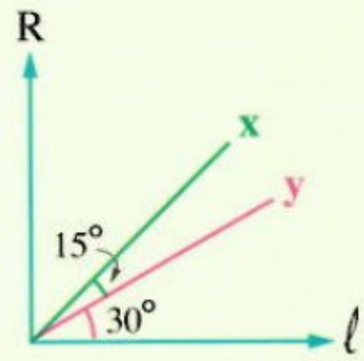
(a) Figures (1) and (3)
(b) Figures (2) and (4)
(c) Figures (1) and (4)
(d) Figures (2) and (3) *(2nd session 24)*

20 A battery of emf V_B that has a negligible internal resistance is connected to a non-insulated resistance wire ab of uniform cross-section, a voltmeter and a sliding contact as shown in the opposite figure, so when moving the sliding contact from point a to b, the reading of the voltmeter ..... . *(Practice 25)*



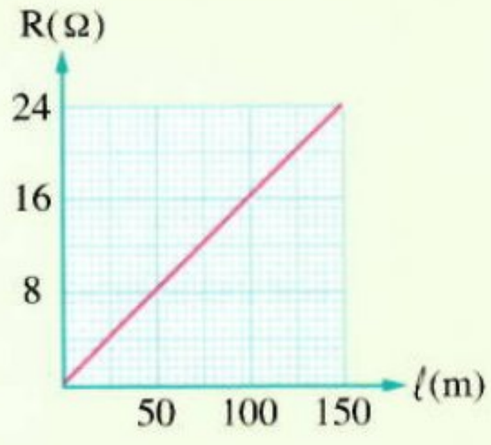
(a) increases
(b) decreases but it doesn't reach zero
(c) remains unchanged
(d) decreases until it reaches zero

21 Two long copper wires x and y are different in thickness, where the opposite graph represents the relation between the resistance (R) and the length (ℓ) taken from each wire, so the ratio between the cross-sectional areas of the two wires $\left(\frac{A_x}{A_y}\right)$ is ..... . *(Practice 25)*



(a) $\frac{1}{3}$
(b) $\frac{3}{1}$
(c) $\frac{1}{\sqrt{3}}$
(d) $\frac{\sqrt{3}}{1}$

22 The opposite graph represents the relation between the electric resistance (R) of each wire of a group of wires made of the same material and having the same cross-sectional area that is 0.1 cm^2 and length (ℓ) for each of these wires, then the resistivity of the material of these wires (ρ_e) equals ..... . *(Practice 25)*



(a) 2.4×10^{-7} Ω.m
(b) 3.6×10^{-7} Ω.m
(c) 1.2×10^{-6} Ω.m
(d) 1.6×10^{-6} Ω.m

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23 Which of the following changes leads to an increase in the electrical resistance of a metal wire?

(1st session 25)

(a) Reducing the wire's cross-sectional area. (b) Lowering the wire's temperature.

(c) Decreasing the wire's length. (d) Coating the wire with an insulating layer.

---

24 In the following figures, three wires (X, Y, and Z) of the same metal have lengths (ℓ, 2 ℓ, and ℓ), and cross-sectional areas (2A, A, and A), respectively?



Wire X Wire Y Wire Z

If the resistance of wire (X) is equal to R, then the resistances of the two wires (Y) and (Z) are ..... .

(2nd session 25)

	Resistance of wire Y	Resistance of wire Z
(a)	2 R	4 R
(b)	8 R	4 R
(c)	R	2 R
(d)	4 R	2 R

---

25 An electric conductor of length 1 m, cross-sectional area 2 mm^2, carries an electric current of intensity 2 A, when the potential difference across its terminals equals 6 V, so this means that ..... .

(Practice 26)

(a) the electric power consumed by the conductor is 3 W

(b) the resistivity of the conductor material is 6×10^6 Ω.m

(c) the conductivity of the conductor material is $1.667 \times 10^5\ \Omega^{-1}.m^{-1}$

(d) the resistance of the conductor is 12 Ω

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